Heinrich Heine
University Düsseldorf

Polymer-Biomolecule Conjugates

Polymer-Biomolecule Conjugates

- Basic Concepts -

Why Polymer-Biomolecule Conjugates?

Synthetic polymers

- lead to an infinite chemical diversity
- can provide new features to the biomolecules while maintaining the primary properties of the latter:
  - + stabilization
  - + protection against degradation
  - + modulation or even enhancement of activity
  - + new functionality

Biomolecules

- may provide means to self-assemble polymers, therefore chemical entities
- Possess the most advanced degree of definition (sequence), therefore of organization

Why Polymer-Biomolecule Conjugates?

PEGylation

Biological active molecules require a recognition of delivery system to be introduced into specific cells or guided to a specific site in the organism. Often, the biologically active molecule needs to be protected from degradation while circulating in the body.

In addition, some peptide drugs suffer from solubility issues.

"The attachment of polyethylene glycol (PEG) to protein or peptide therapeutics, termed PEGylation, is an example of a highly successful strategy that gives rise to several benefits including:
- increased bioavailability and plasma half-lives,
- decreased immunogenicity
- reduced proteolysis
- enhanced solubility and stability."*

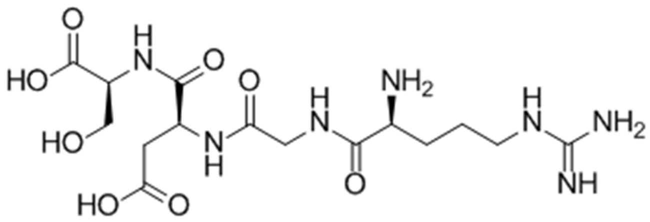
Polymer-Biomolecule Conjugates

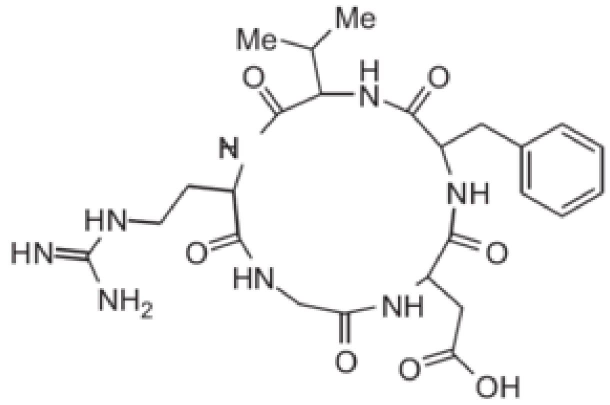
- Polymer-Protein Conjugates -

Biomolecules

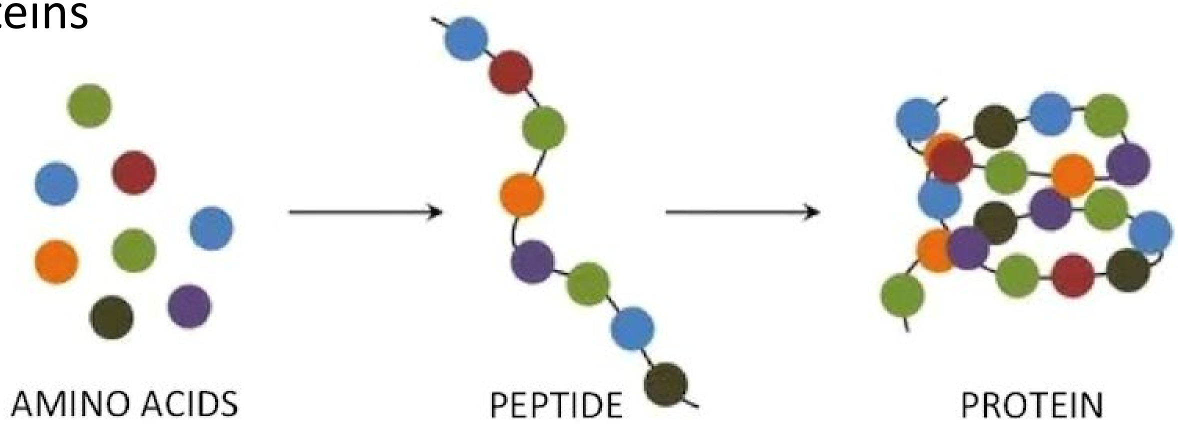
Main species

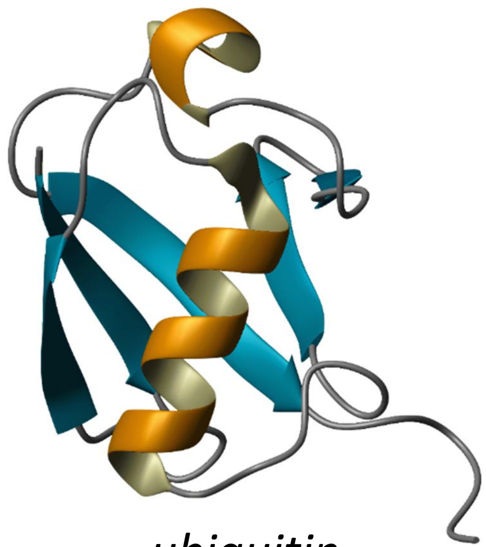
Peptides





Proteins



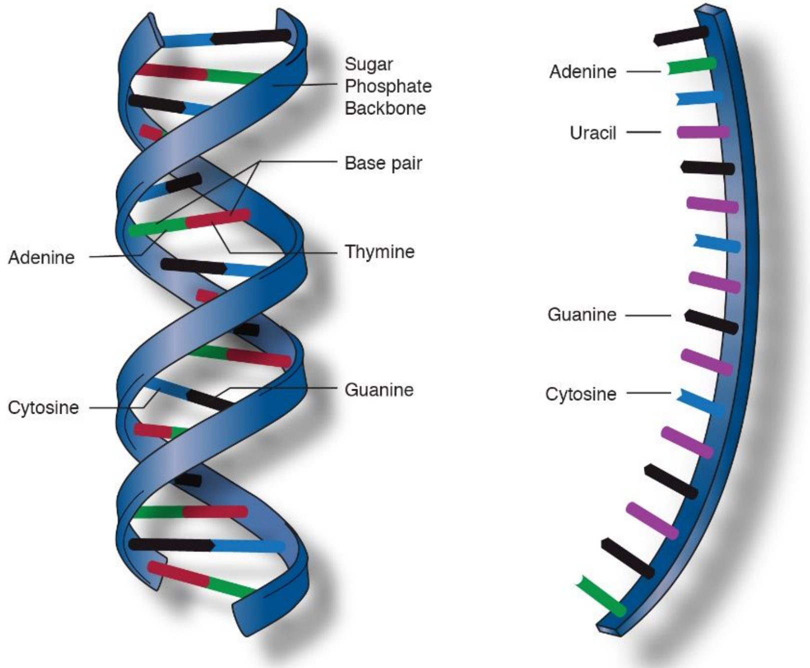


ubiquitin

Biomolecules

Main species

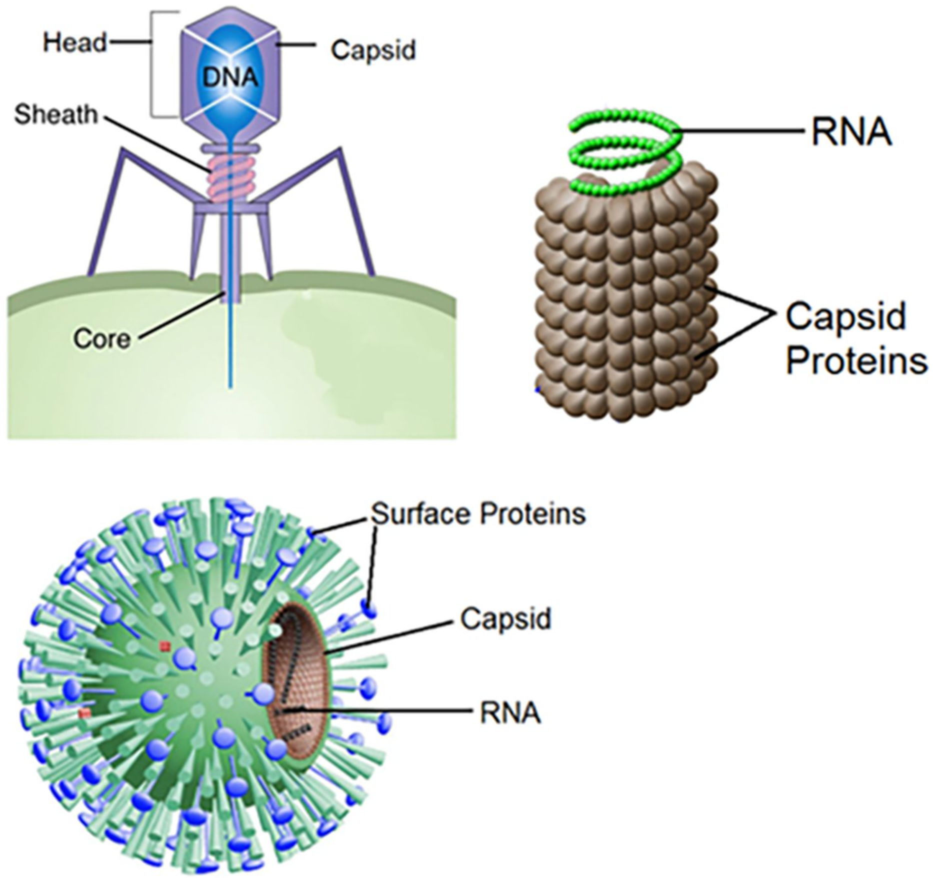
Nucleic acids (e.g., DNA, RNA)



Deoxyribonucleic acid (DNA)

Ribonucleic acid (RNA)

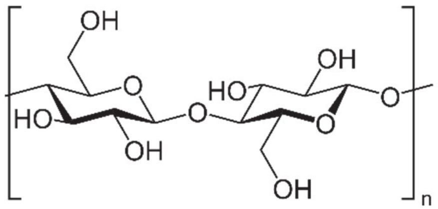
Viruses



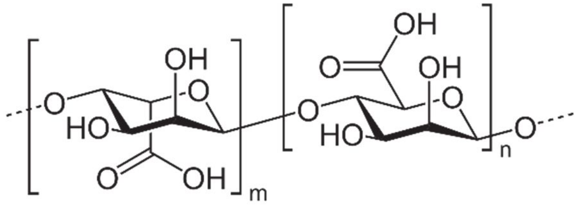
Biomolecules

Main species

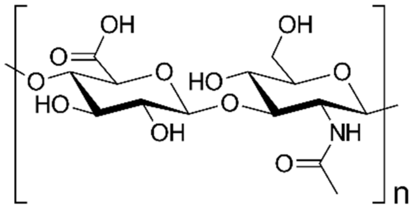
Polysaccharides



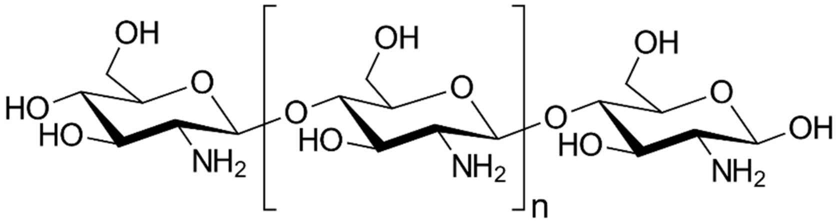
cellulose



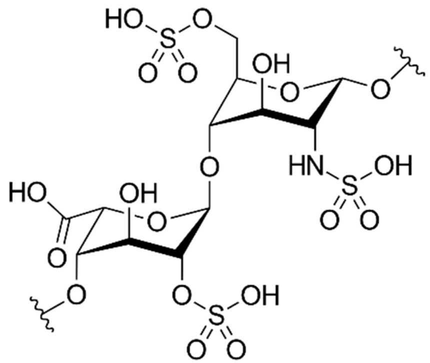
alginic acid



hyaluronic acid



chitosan

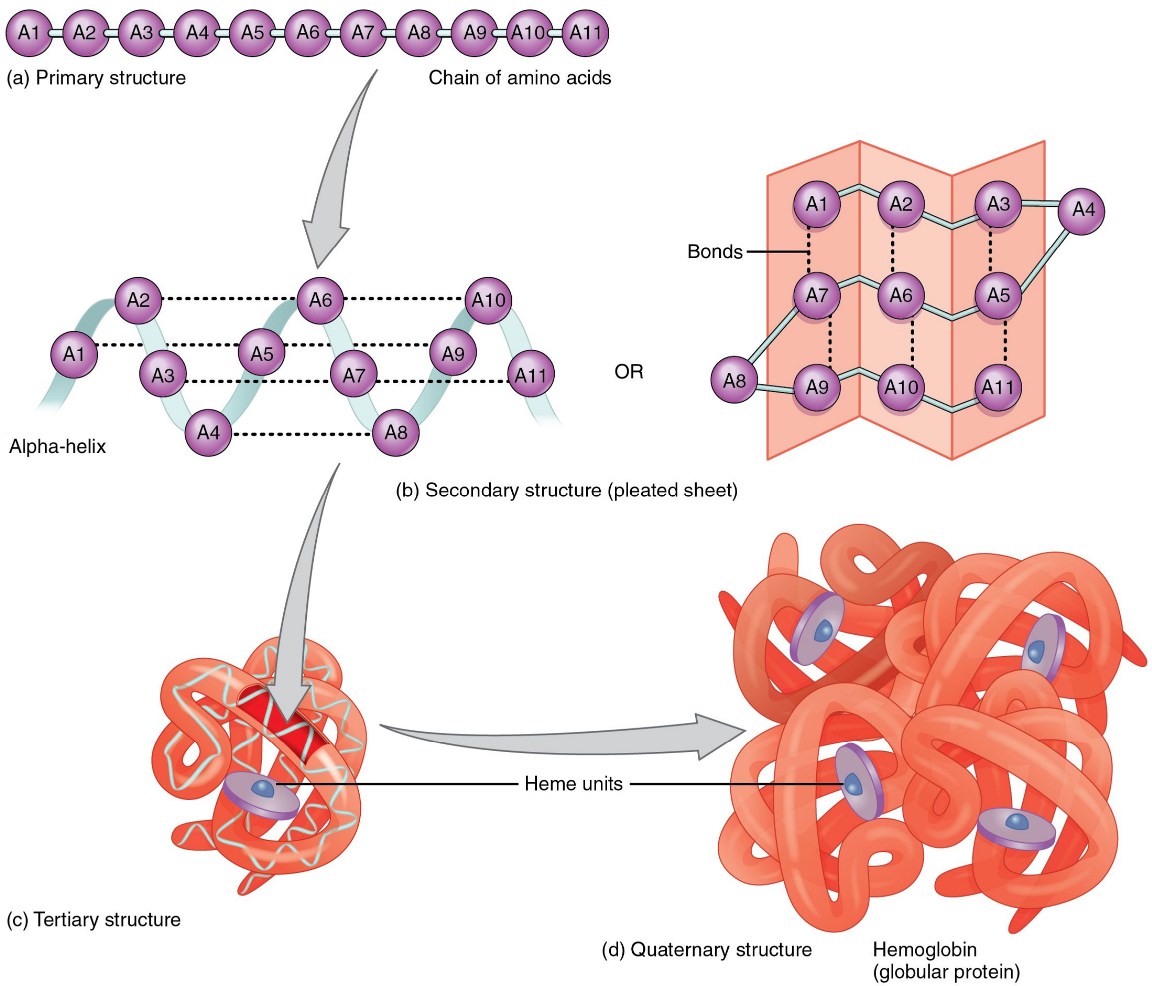


heparin

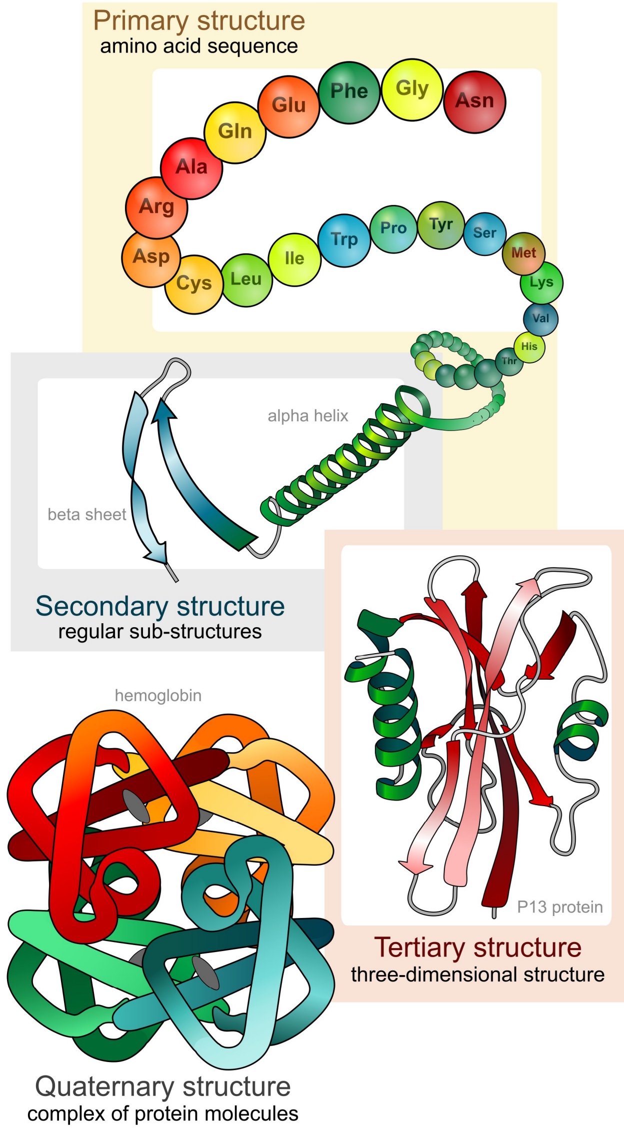
Structure/Nature of Biomolecules

Proteins

Primary structure: sequence of amino acids



faculty.fmcc.suny.edu

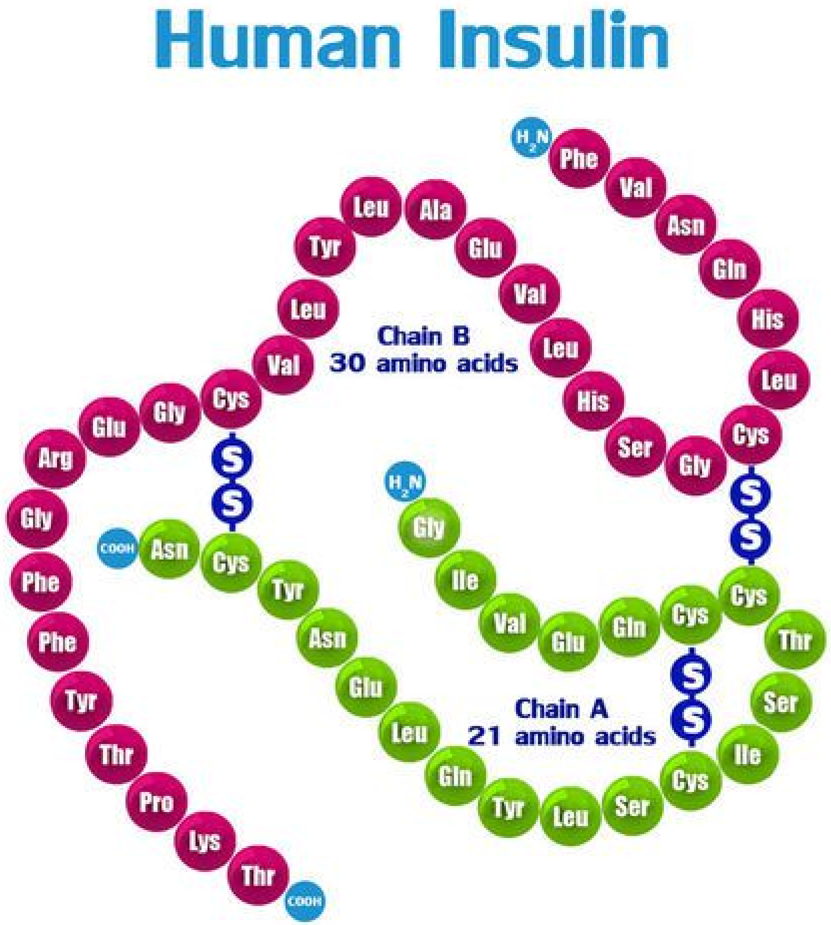


en.wikipedia.org “protein structure”

Structure/Nature of Biomolecules

Proteins

Primary structure: sequence of amino acids



Structure/Nature of Biomolecules

Proteins

Secondary structure: local sub-structures (α-helices, β-sheets)

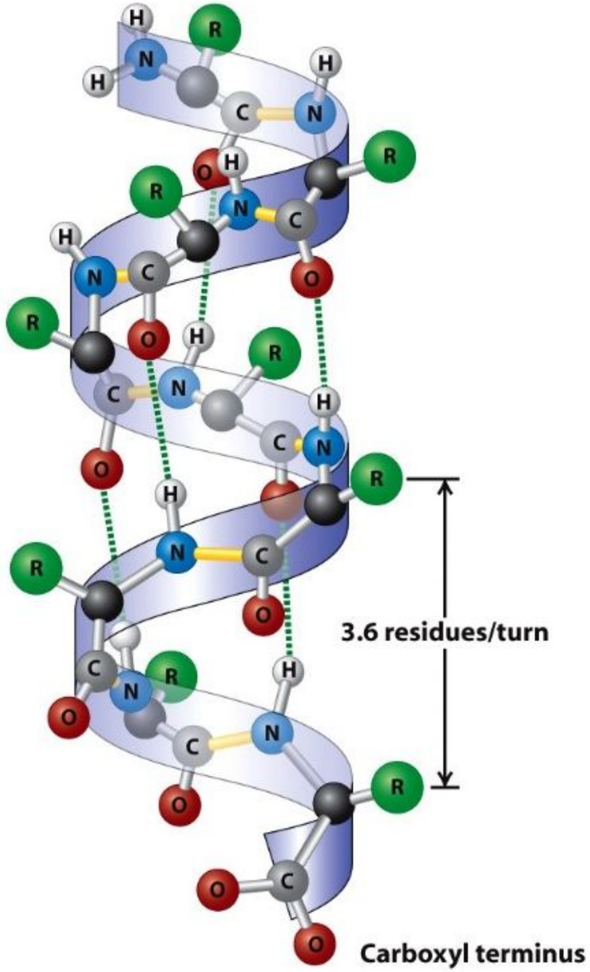


Figure 3-4
Molecular Cell Biology, Sixth Edition

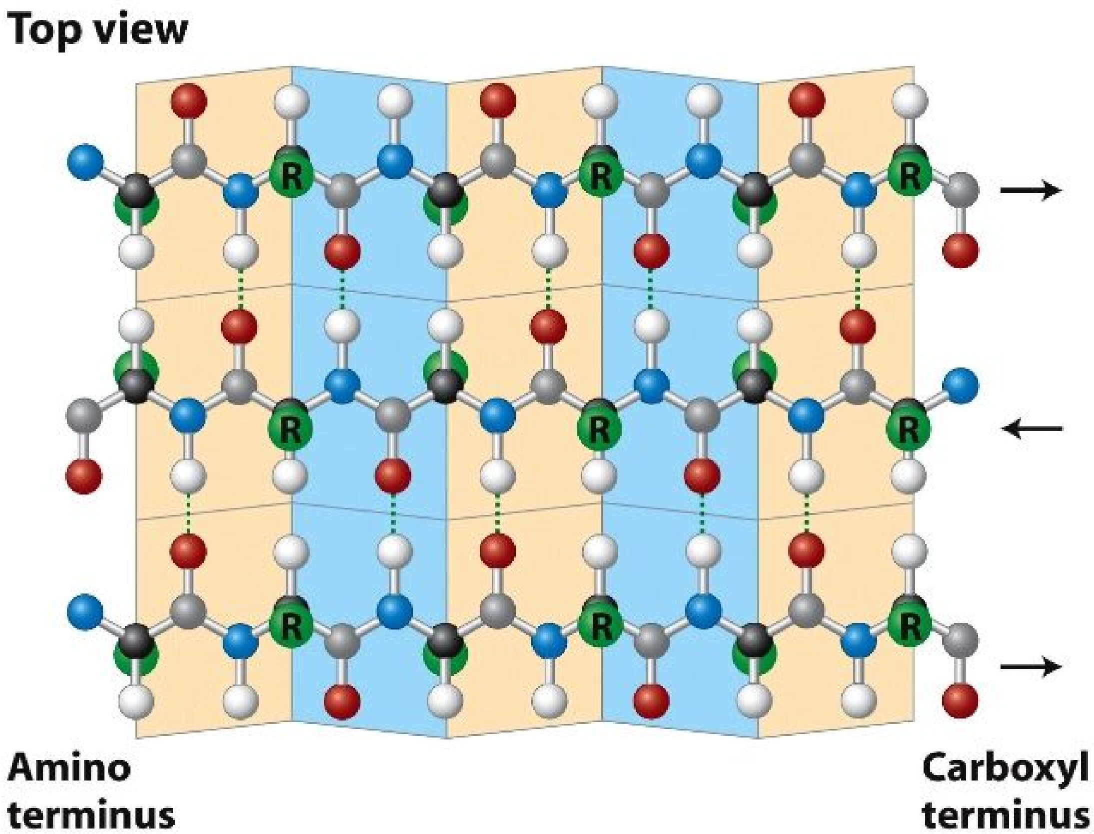


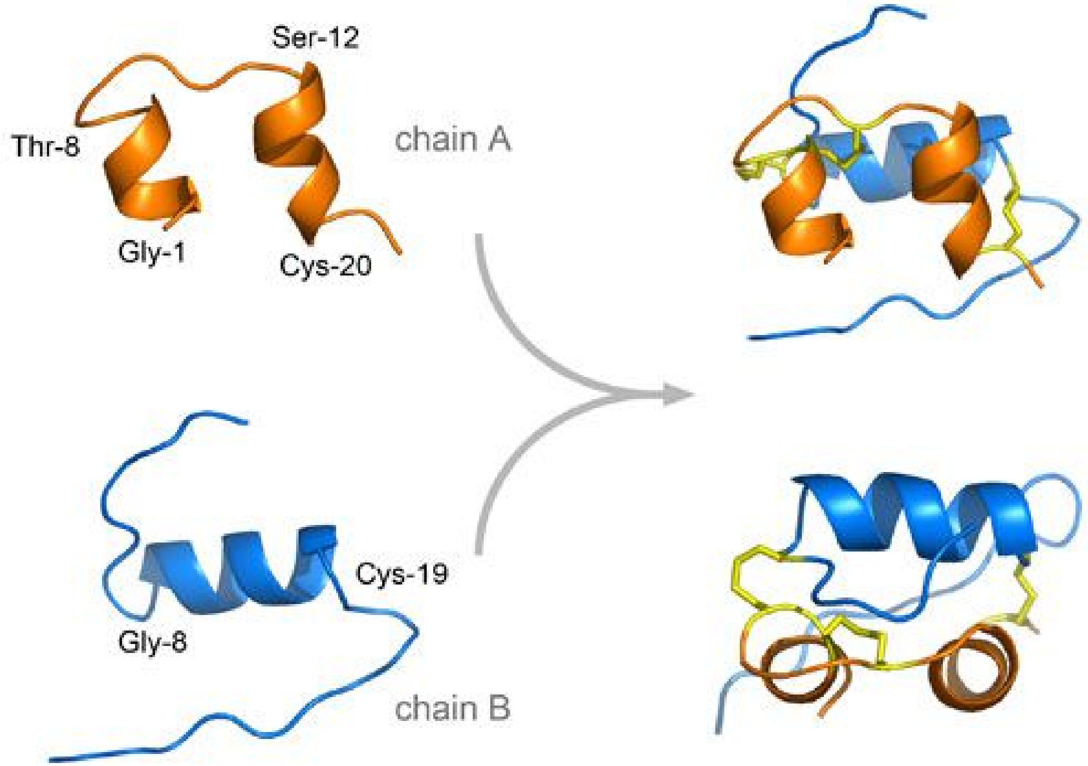
Figure 3-5a
Molecular Cell Biology, Sixth Edition

Structure/Nature of Biomolecules

Proteins

Tertiary structure: assembly of all secondary structures (folding) to bury hydrophobic parts and expose polar residues

Insulin

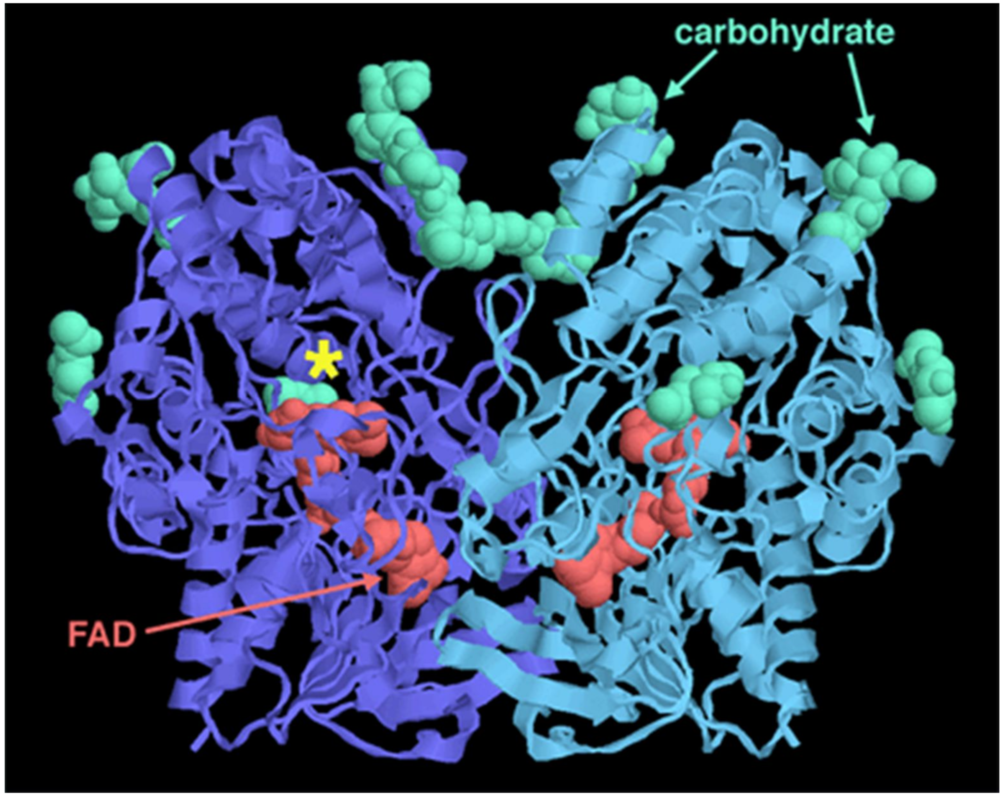


Structure/Nature of Biomolecules

Proteins

Quaternary structure: three-dimensional structure of a multi-subunit protein

Glucose Oxidase



Any alteration to any of these structural levels may alter activity! + or -

Structure/Nature of Biomolecules

Proteins

Four structural levels

TABLE 3.2 A Summary of Protein Structure

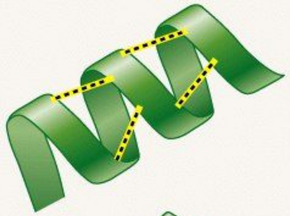
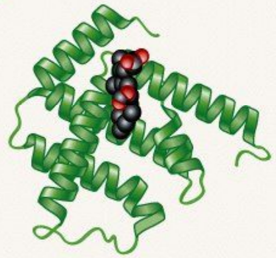
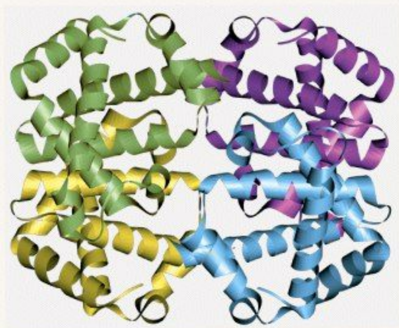
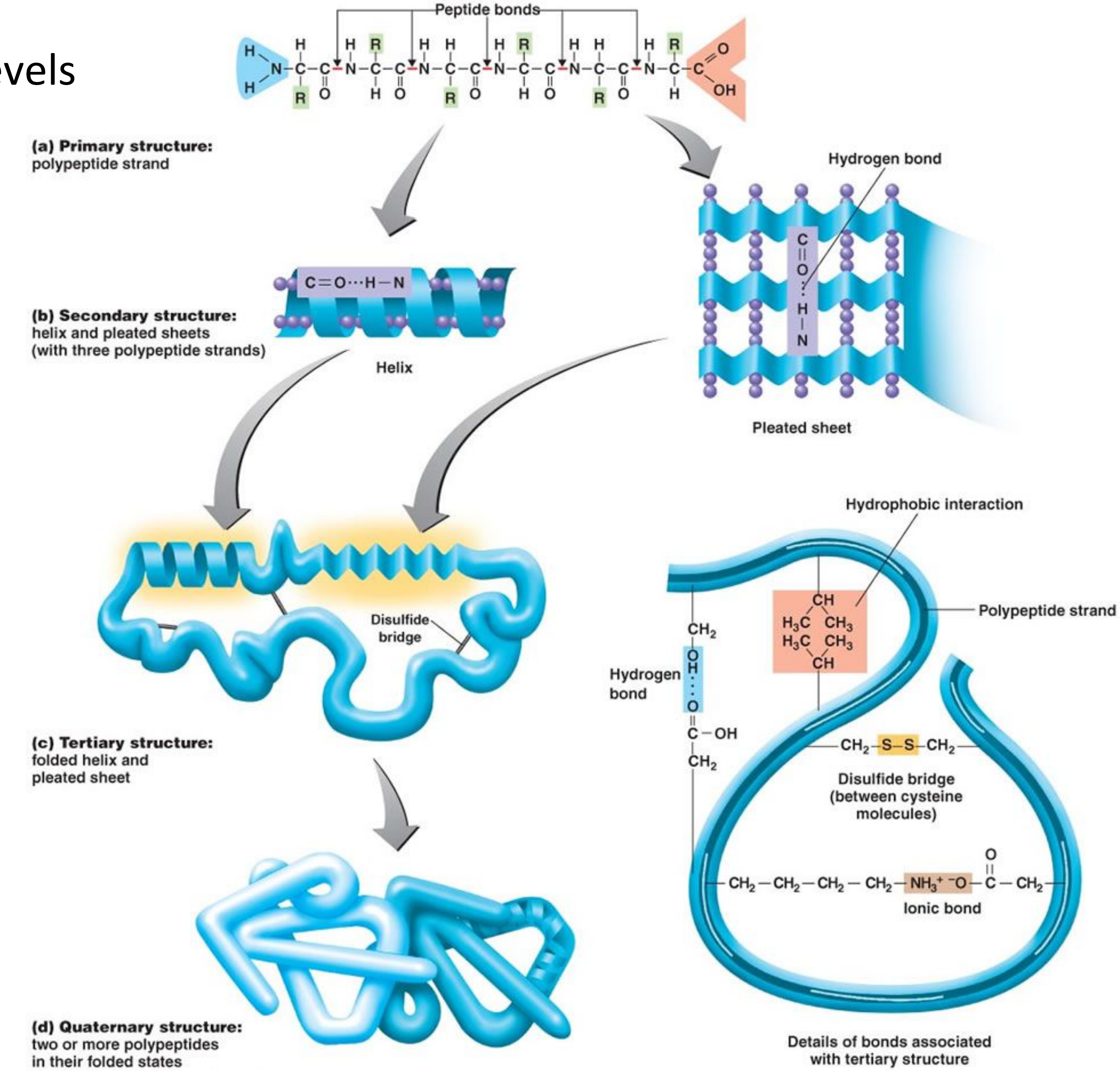
Level	Description	Stabilized by	Example: Hemoglobin
Primary	The sequence of amino acids in a polypeptide	Peptide bonds	Gly – Ser – Asp – Cys
Secondary	Formation of α-helices and β-pleated sheets in a polypeptide	Hydrogen bonding between groups along the peptide-bonded backbone	
Tertiary	Overall three-dimensional shape of a polypeptide (The model on the right shows one of hemoglobin's subunits. The black and red atoms are in the heme group that carries oxygen; they are not part of the protein itself.)	Bonds and other interactions between R-groups, or between R-groups and the peptide-bonded backbone	
Quaternary	Shape produced by combinations of polypeptides. (The model on the right shows hemoglobin, which consists of four polypeptides.)	Bonds and other interactions between R-groups, and between peptide backbones of different polypeptides	

Table 3-2 Biological Science, 2/e

Structure/Nature of Biomolecules

Proteins

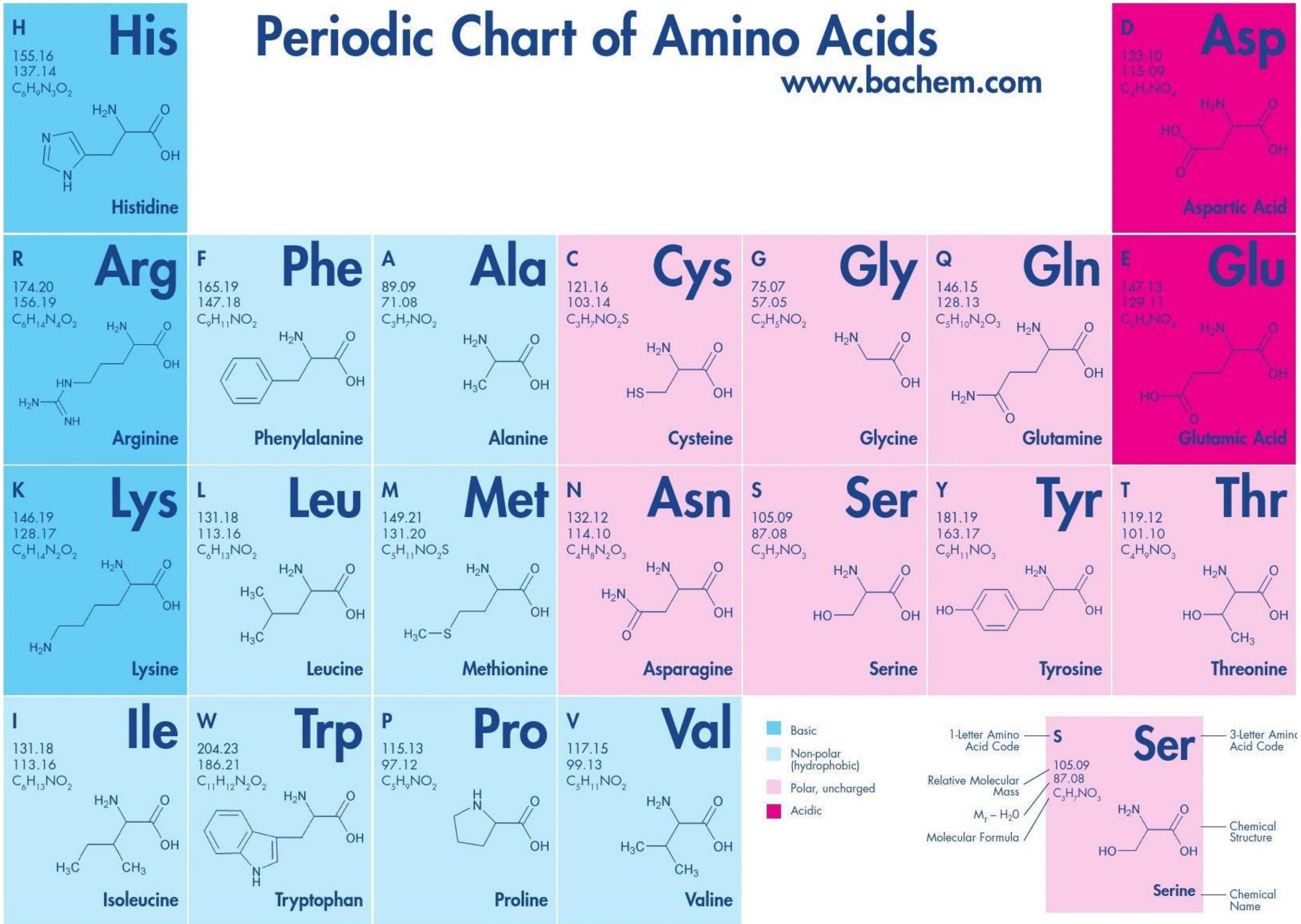
Four structural levels



Structure/Nature of Biomolecules

Proteins – how can we modify them?

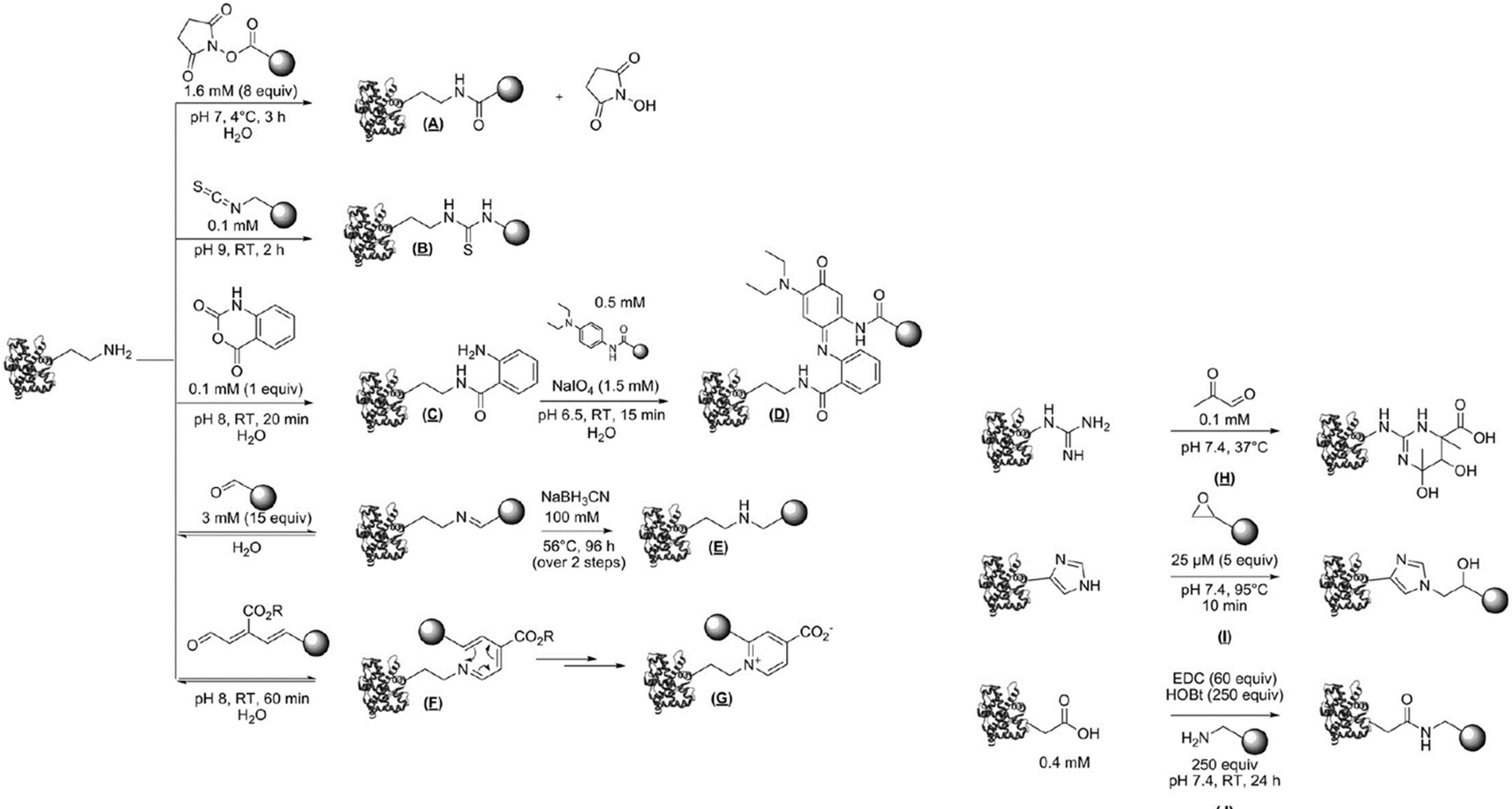
What are the chemical anchors at the surface of a protein?



Modification of Biomolecules

Proteins – how can we modify them?

Targeting canonical amino acids

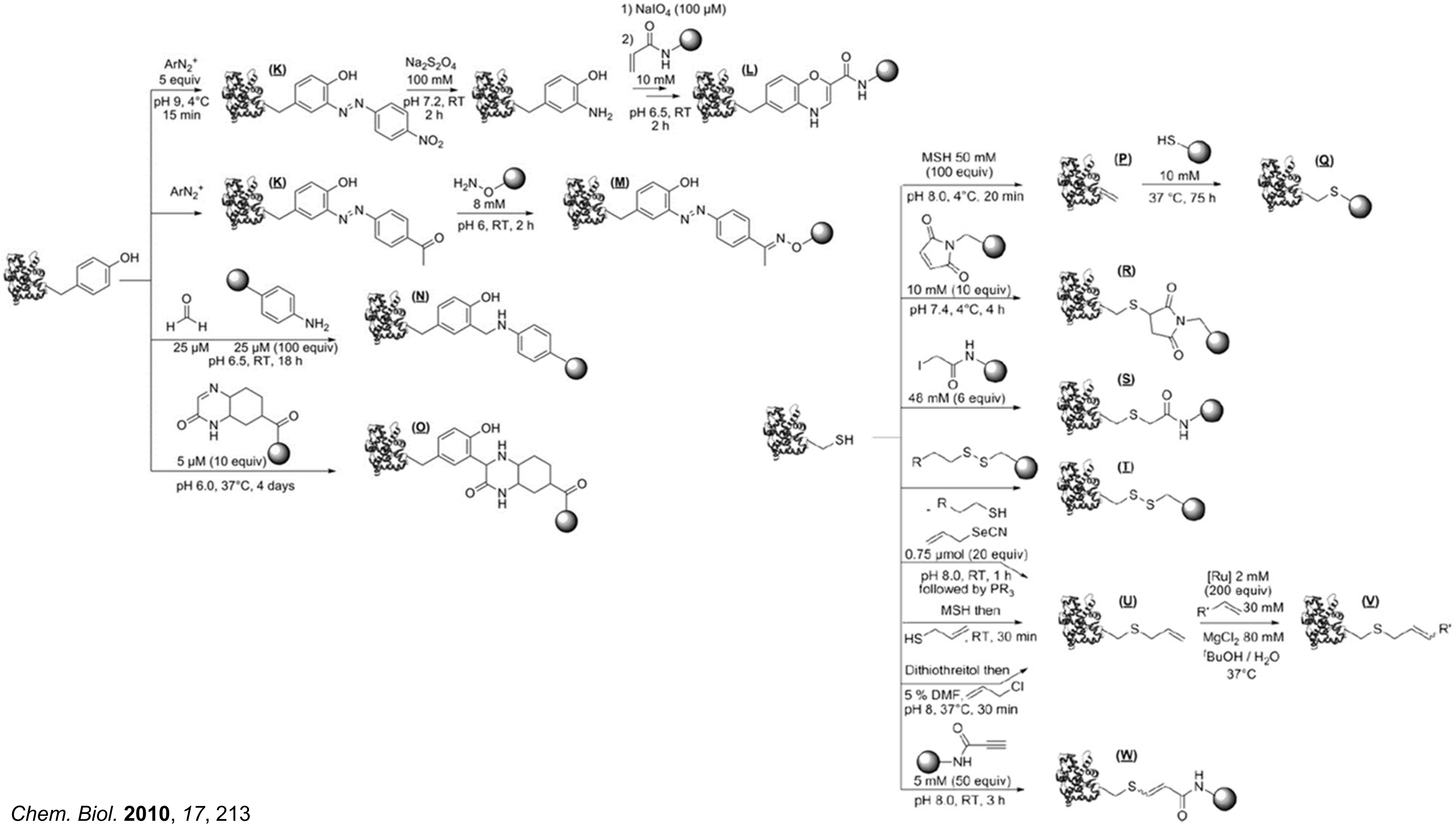


Chem. Biol. **2010**, *17*, 213

Modification of Biomolecules

Proteins – how can we modify them?

Targeting canonical amino acids



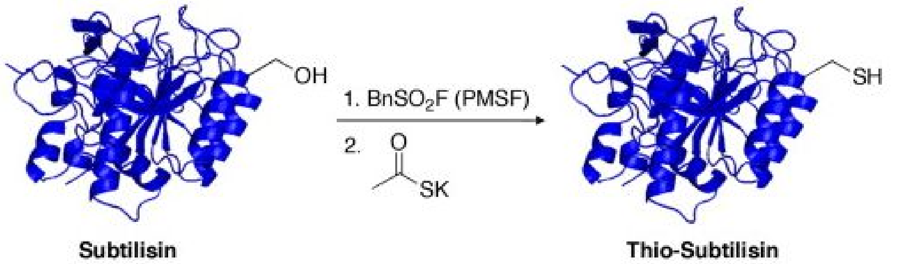
Chem. Biol. **2010**, *17*, 213

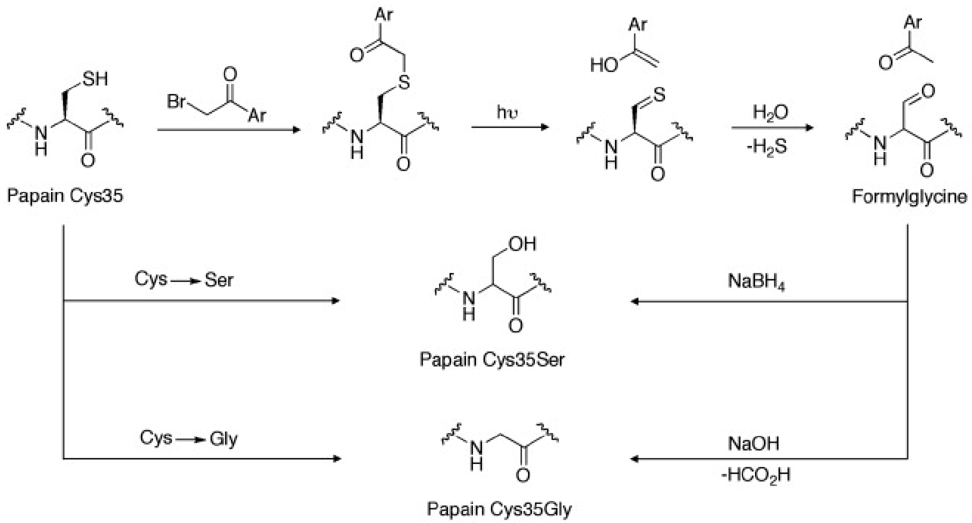
Modification of Biomolecules

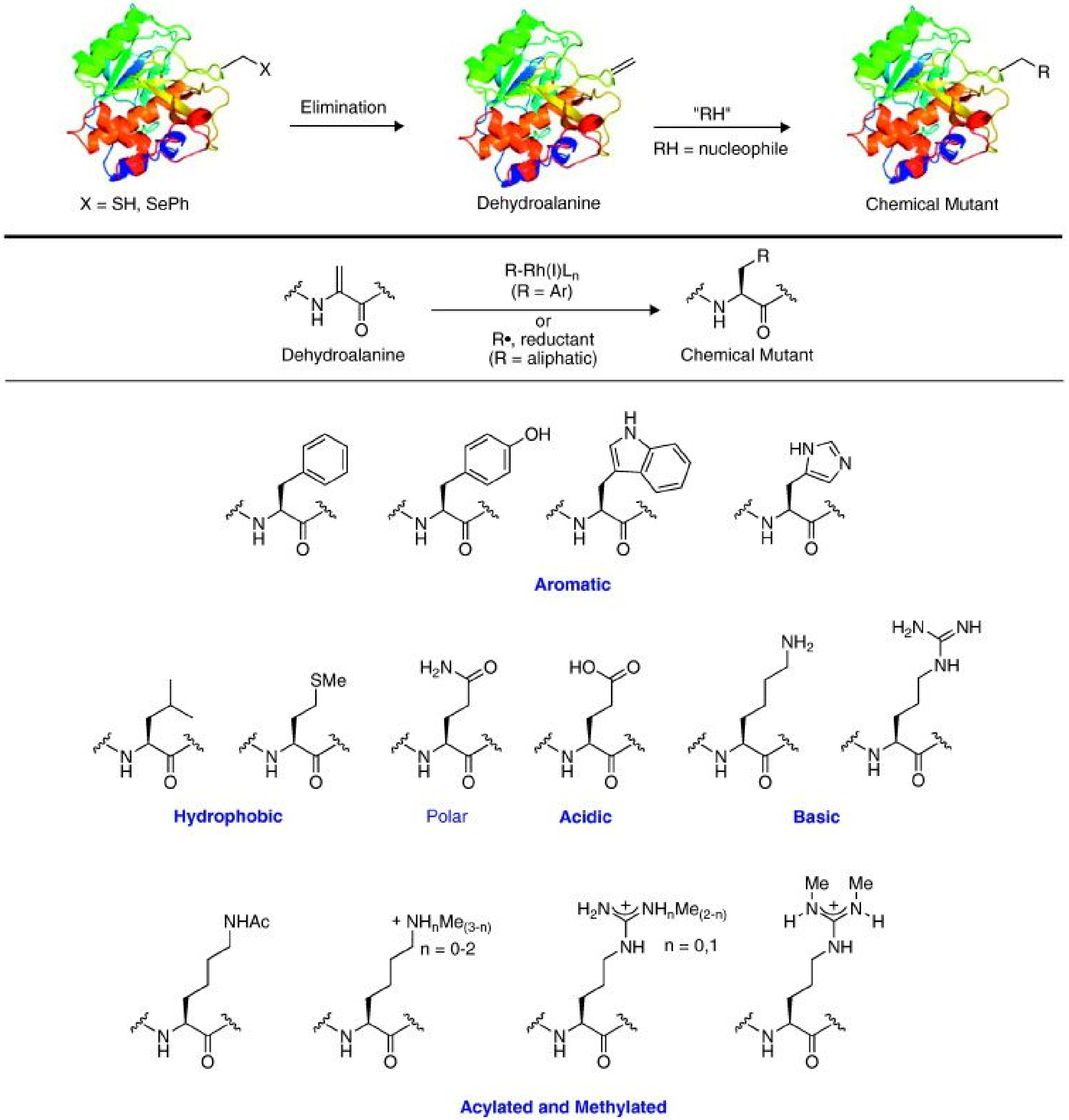
Proteins – how can we modify them?

Targeting canonical amino acids

Chemical mutagenesis
= transformation of an amino acid
into another, directly on the protein







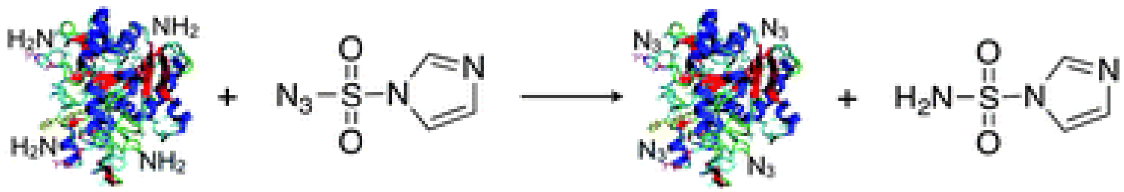
Curr. Opin. Chem. Biol. **2010**, *14*, 781

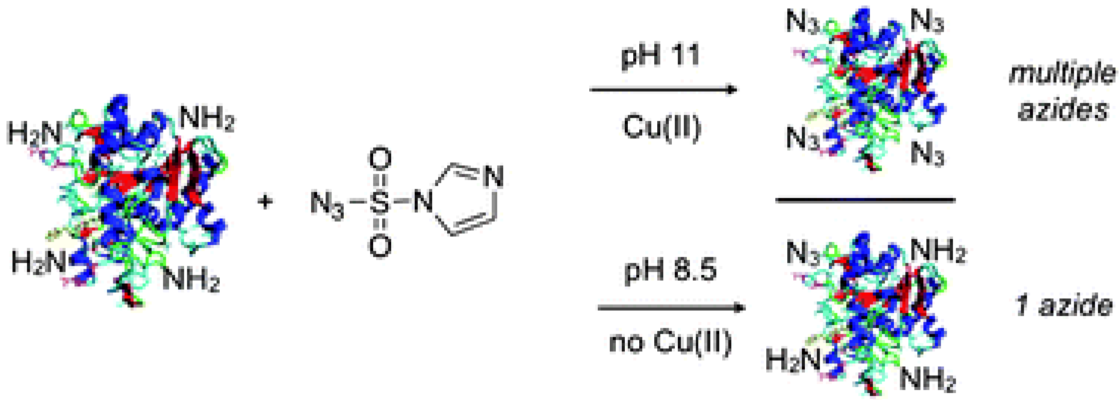
Modification of Biomolecules

Proteins – how can we modify them?

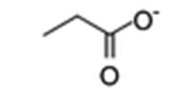
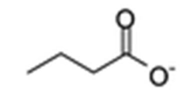
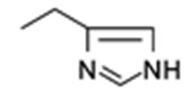
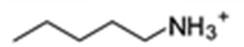
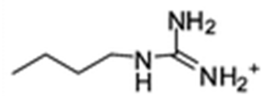
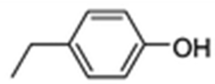
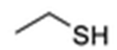
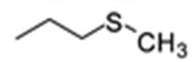
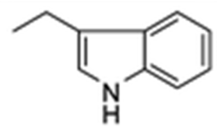
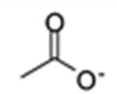
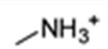
Targeting canonical amino acids

Chemical mutagenesis





Besides man-made synthetic protocols, enzymes can also be employed for site-specific modification.

Amino acid	Side chain	pKa
Aspartic acid		3.7-4.0
Glutamic acid		4.2-4.5
Histidine		6.7-7.1
Lysine		9.3-9.5
Arginine		>12
Tyrosine		9.7-10.1
Cysteine		8.8-9.1
Methionine		
Tryptophan		
C-terminus		2.1-2.4
N-terminus		7.6-8.0

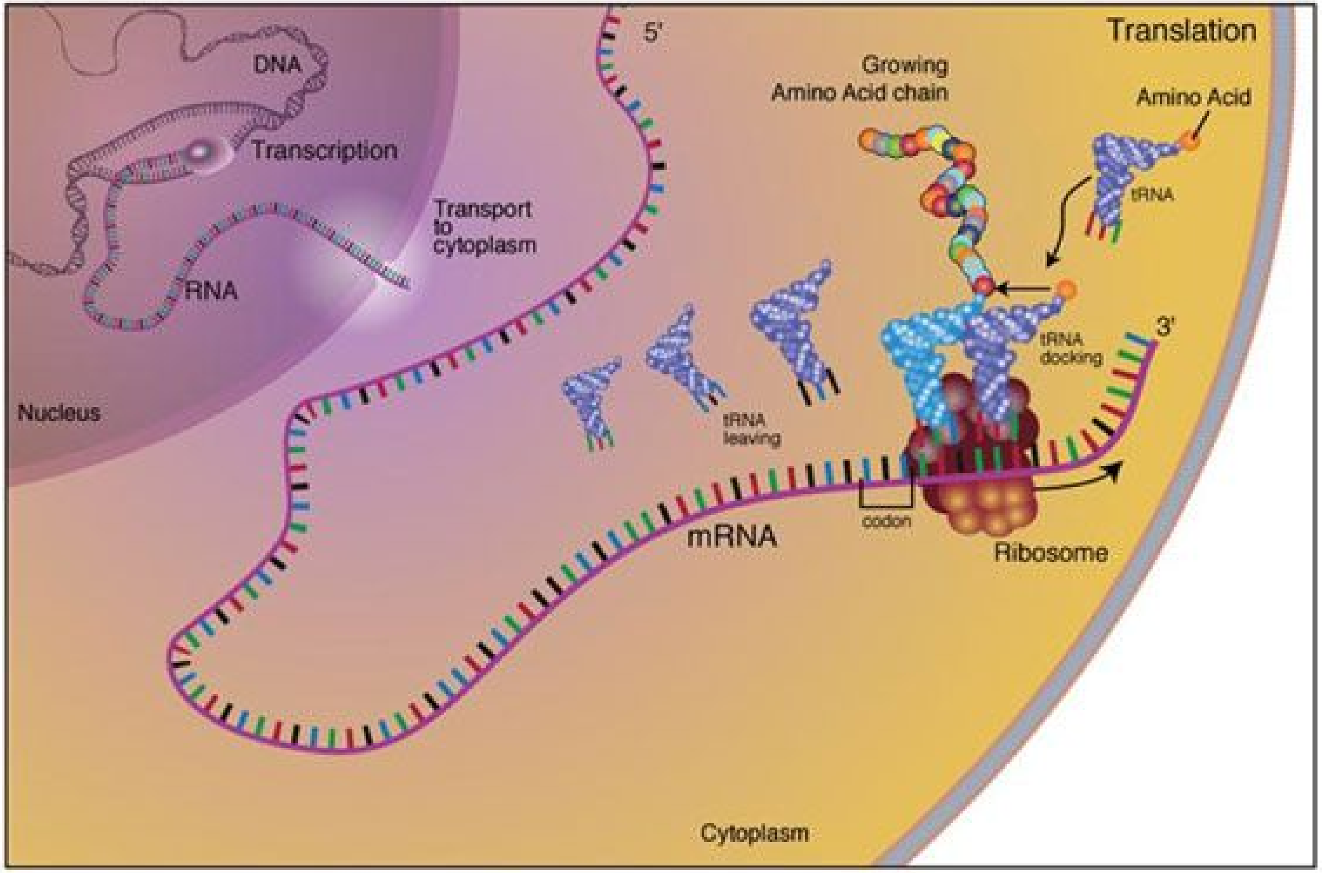
Bioconjug. Chem. **2009**, *20*, 20 ; *Chem. Sci.* **2011**, 2, 701

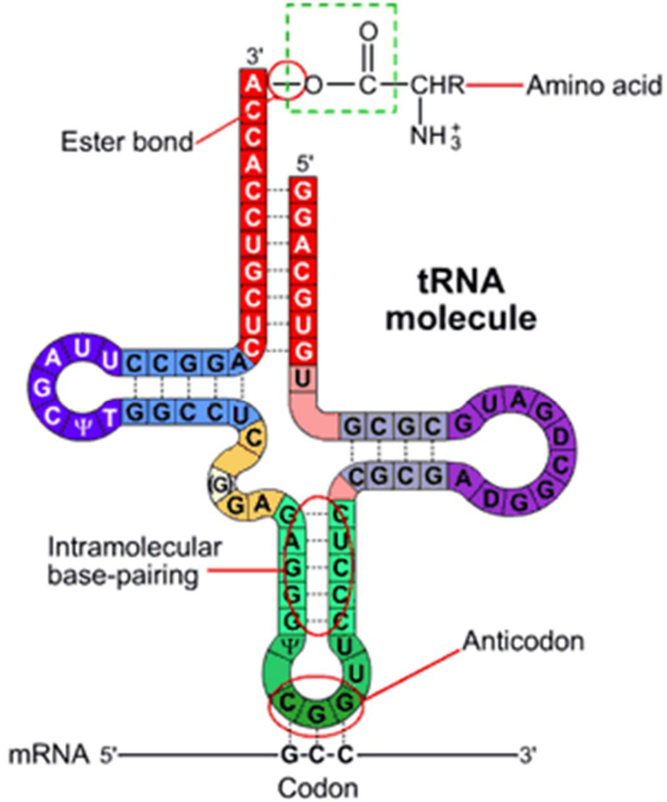
Bioconjug. Chem. **2009**, *20*, 1281

Modification of Biomolecules

Proteins – how can we modify them?

Introduction of non-natural amino acids





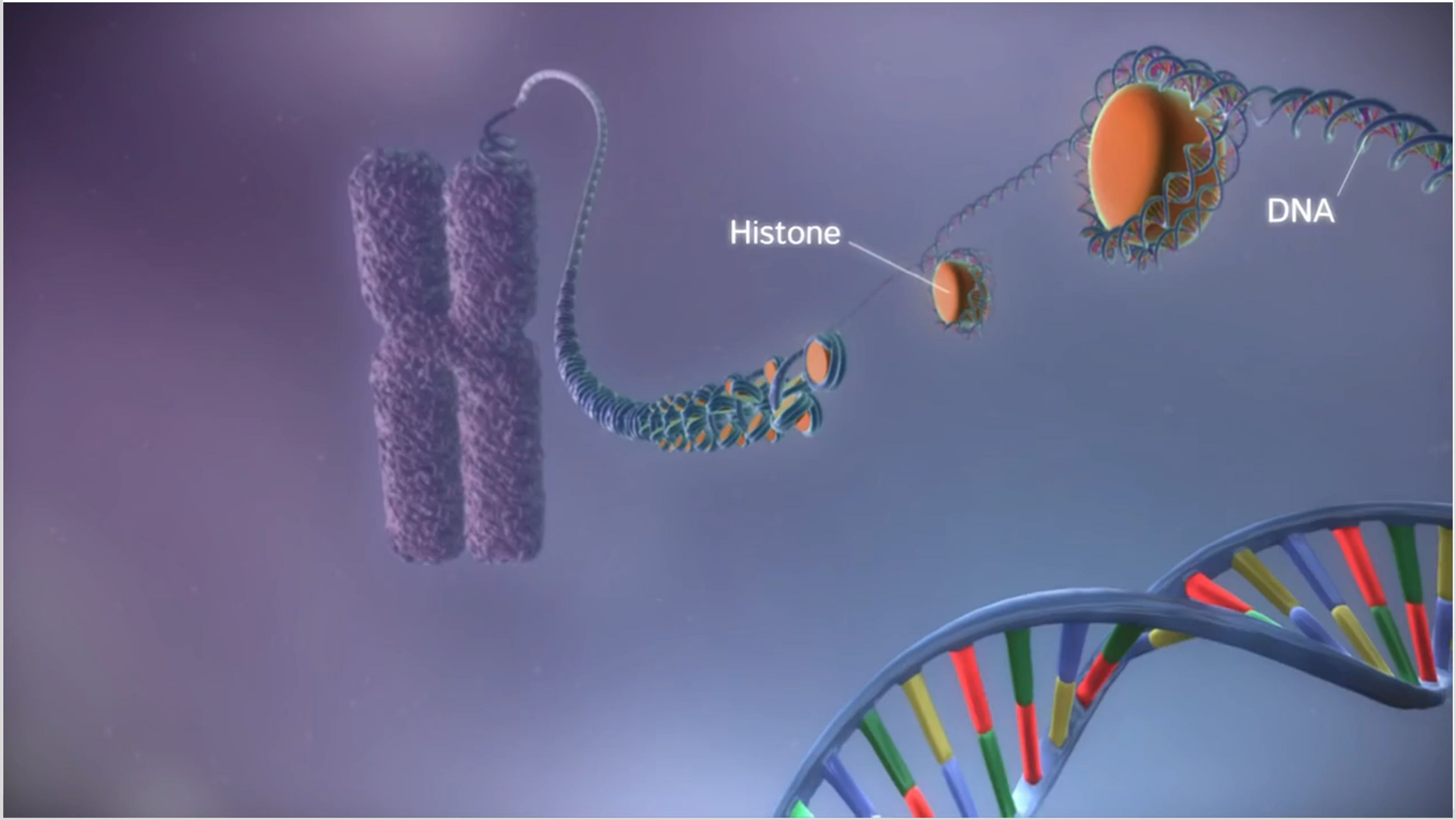
smart-therapeutics.com

proteinsynthesis.org

Modification of Biomolecules

Proteins – how can we modify them?

Brief background on protein synthesis



Modification of Biomolecules

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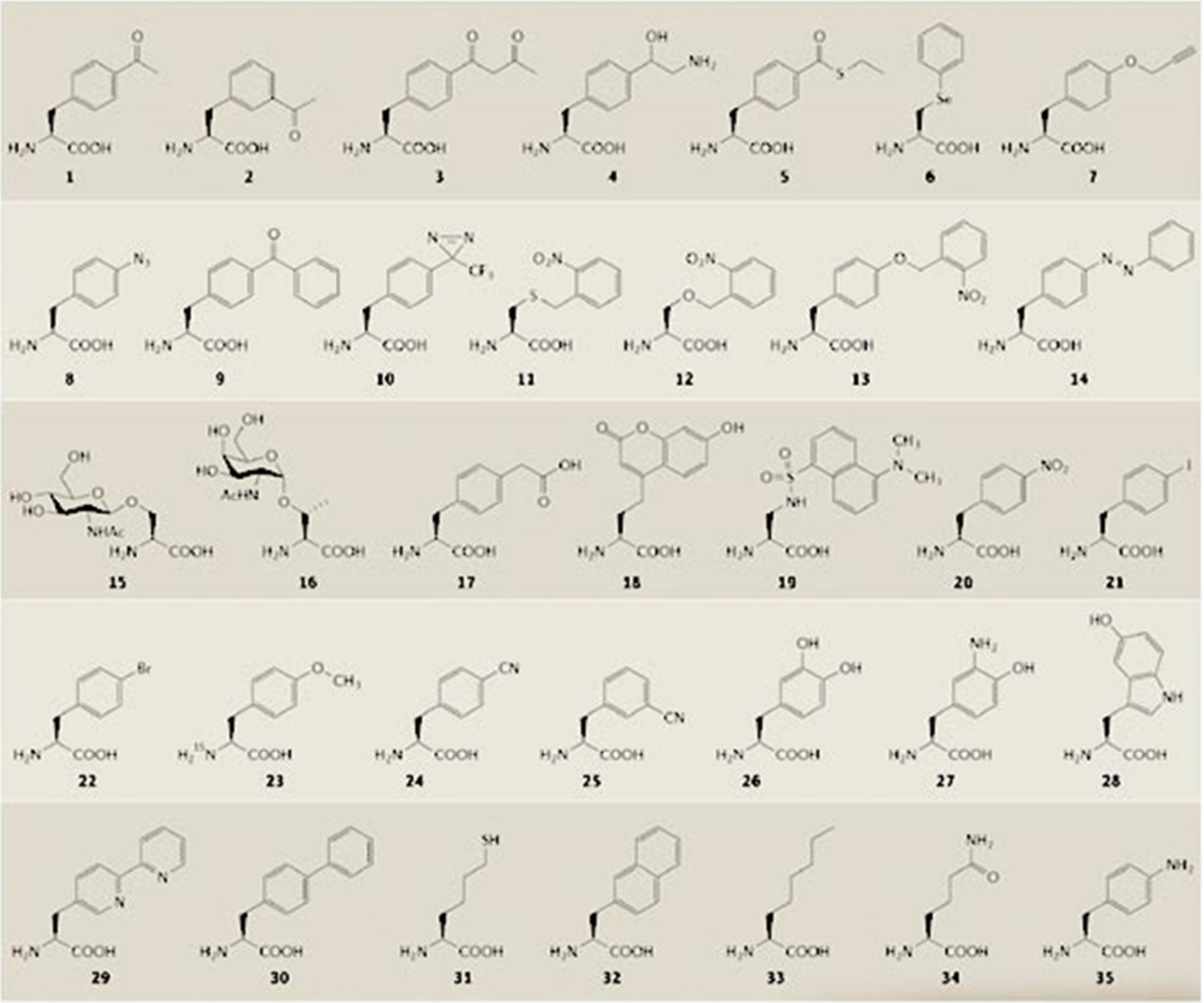
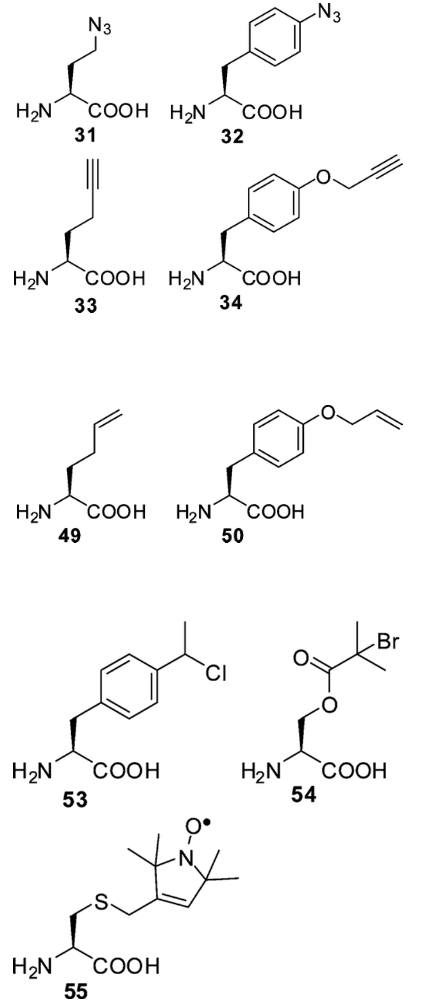




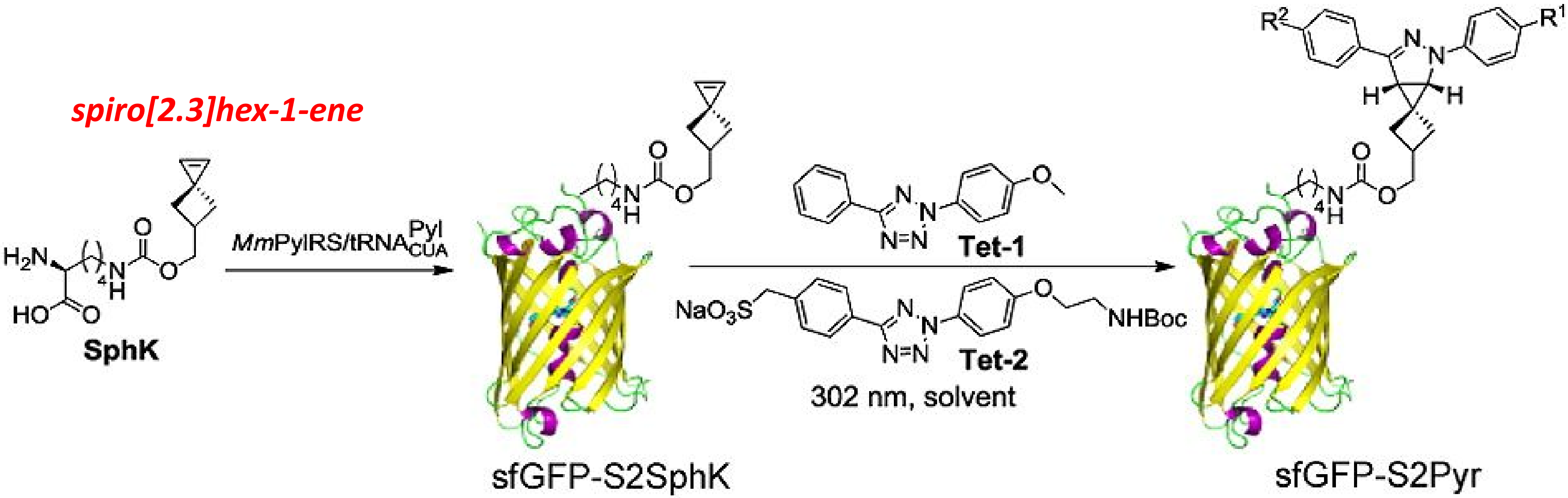
Nat. Rev. Mol. Cell. Biol. **2006**, *7*, 775

Bioconjug. Chem. **2009**, *20*, 1281

Modification of Biomolecules

Proteins – how can we modify them?

Introduction of non-natural amino acids



J. Am. Chem. Soc. **2014**, *136*, 4153

Modification of Biomolecules

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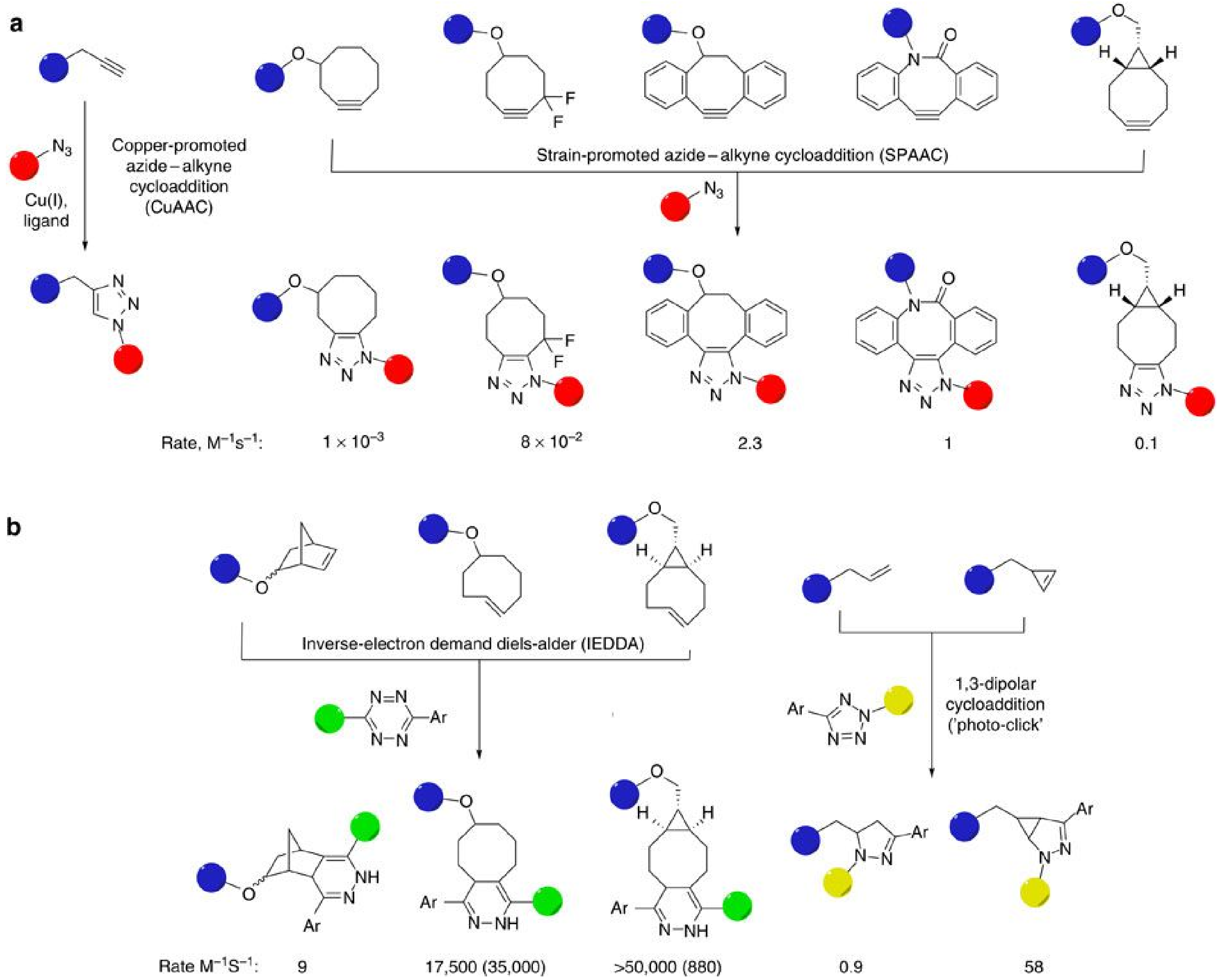

Nature Commun. **2014**, *5*, 4740

Modification of Biomolecules

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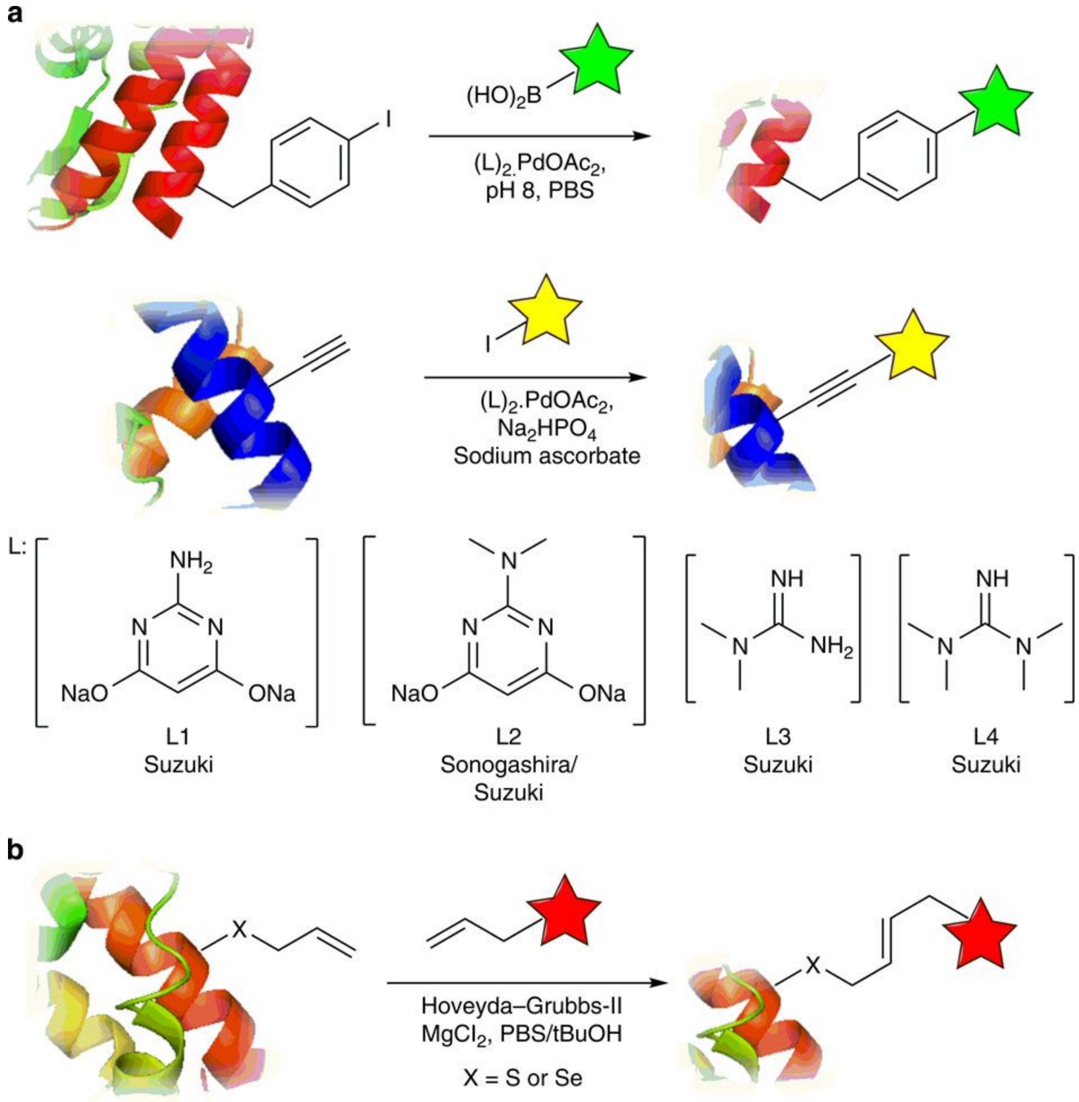


Nature Commun. **2014**, *5*, 4740

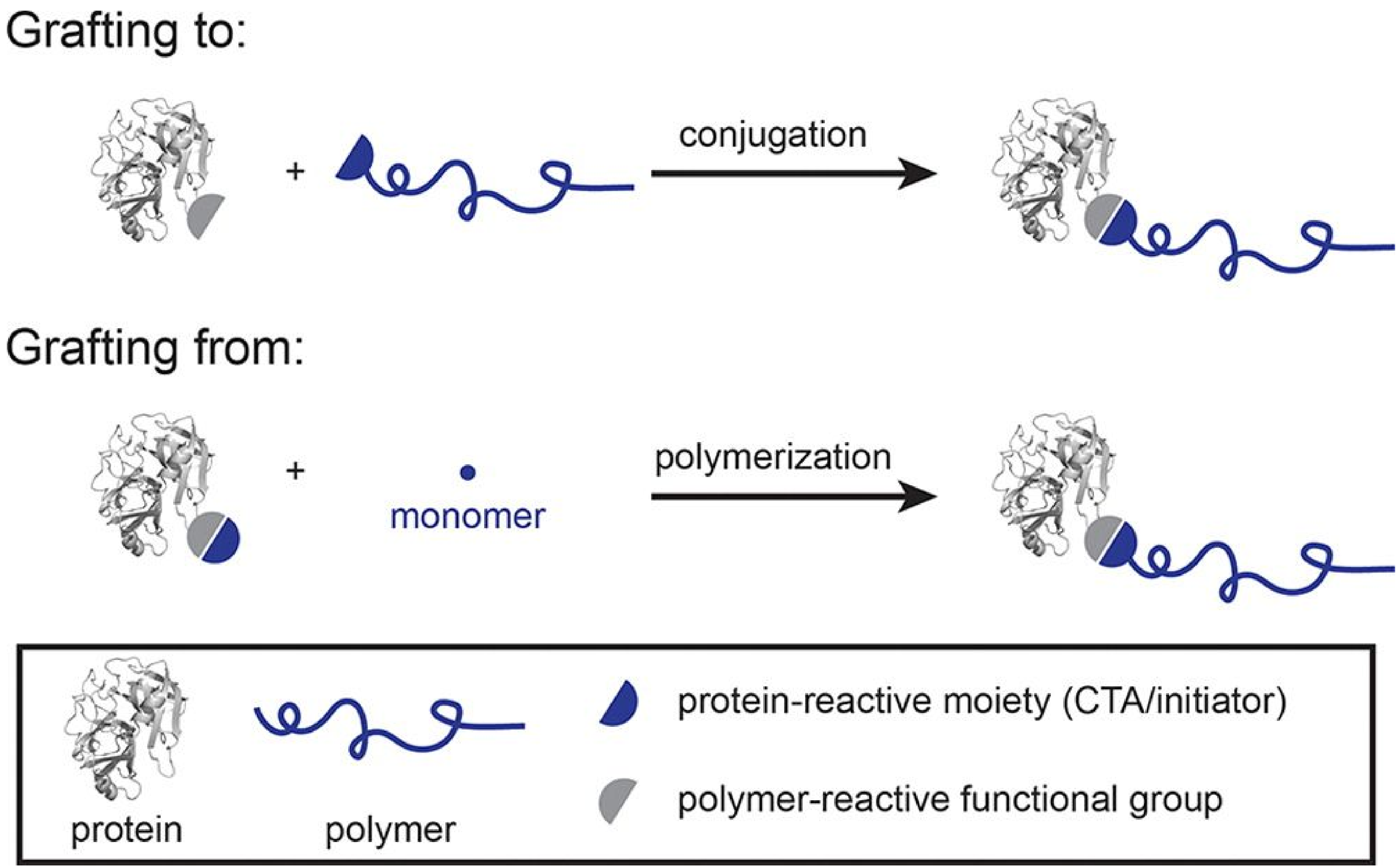
Polymer-Biomolecule Conjugates

- Polymer-Protein Conjugates -

Synthesis

Protein-Polymer Conjugates

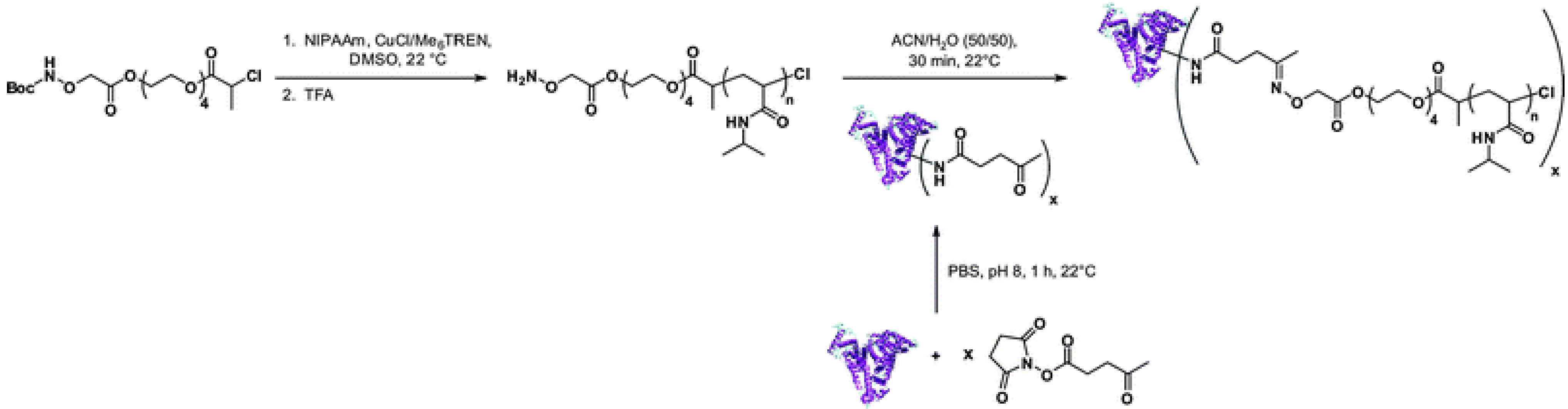
Main synthetic routes



ACS Macro Lett. **2015**, *4*, 101

Protein-Polymer Conjugates

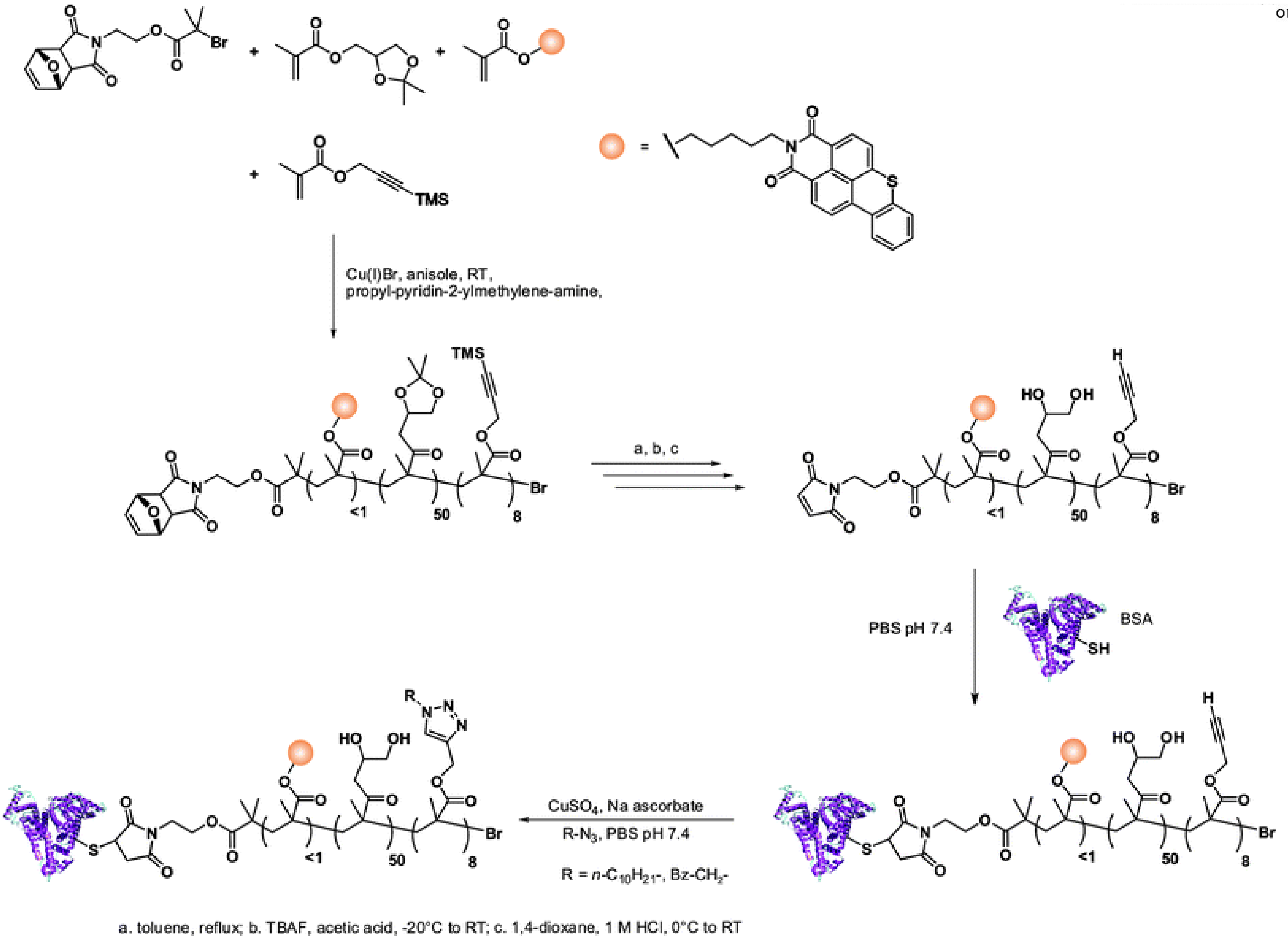
Grafting to



Polym. Chem., **2010**, *1*, 563

Protein-Polymer Conjugates

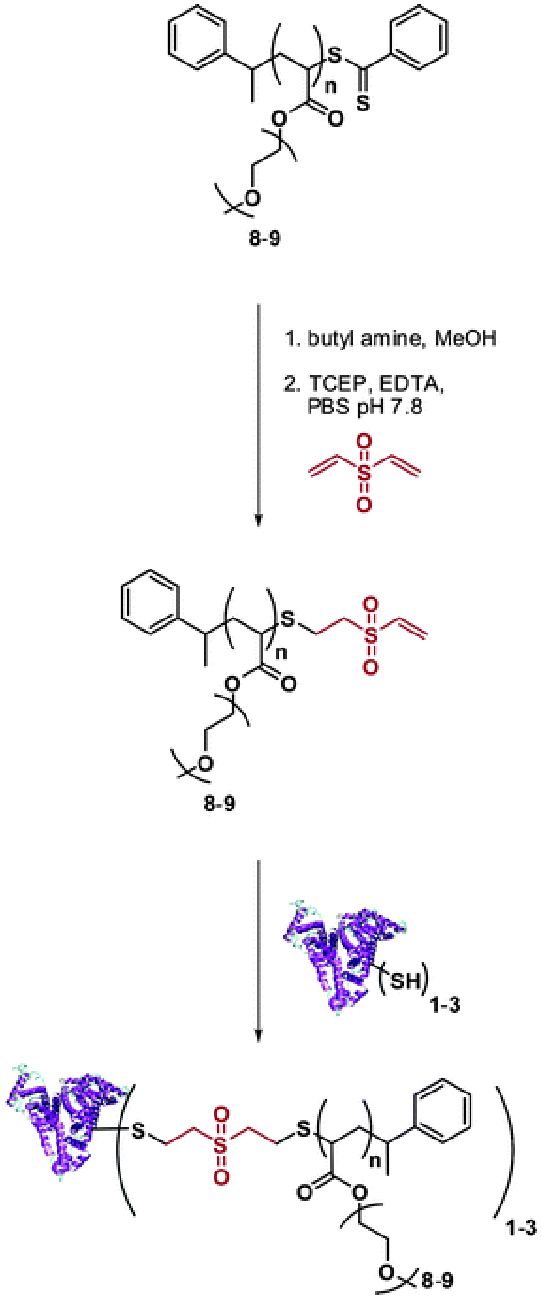
Grafting to



Polym. Chem., **2010**, *1*, 563

Protein-Polymer Conjugates

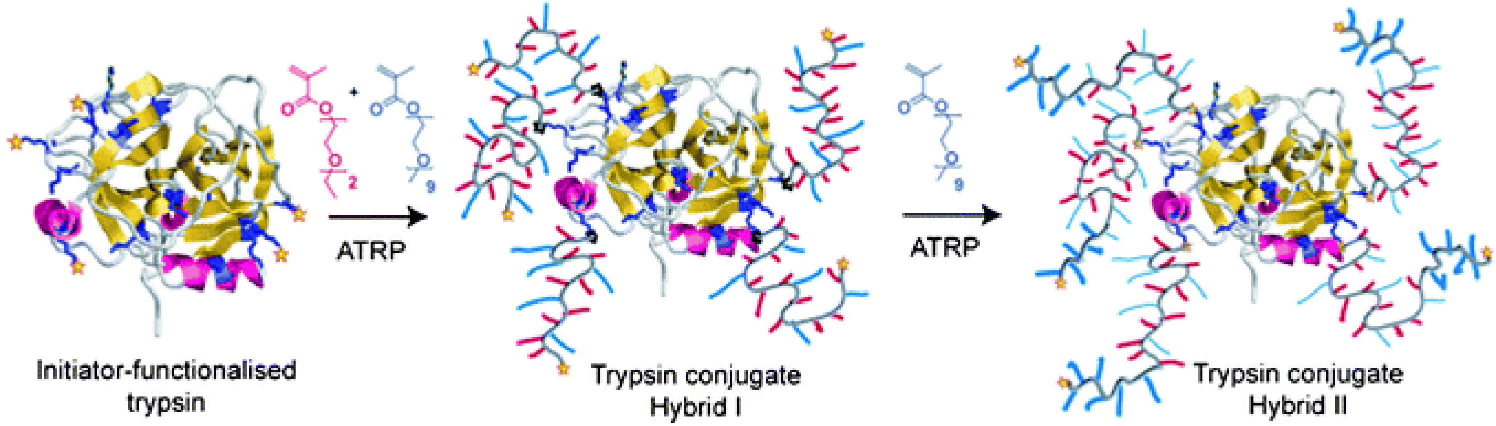
Grafting to

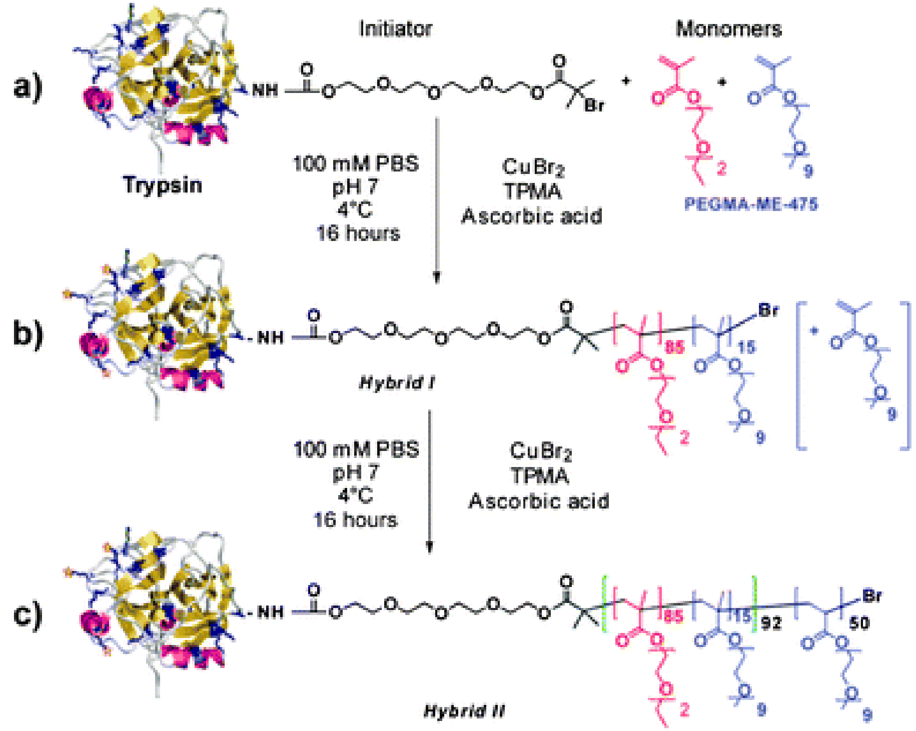


Polym. Chem., **2010**, *1*, 563

Protein-Polymer Conjugates

Grafting from

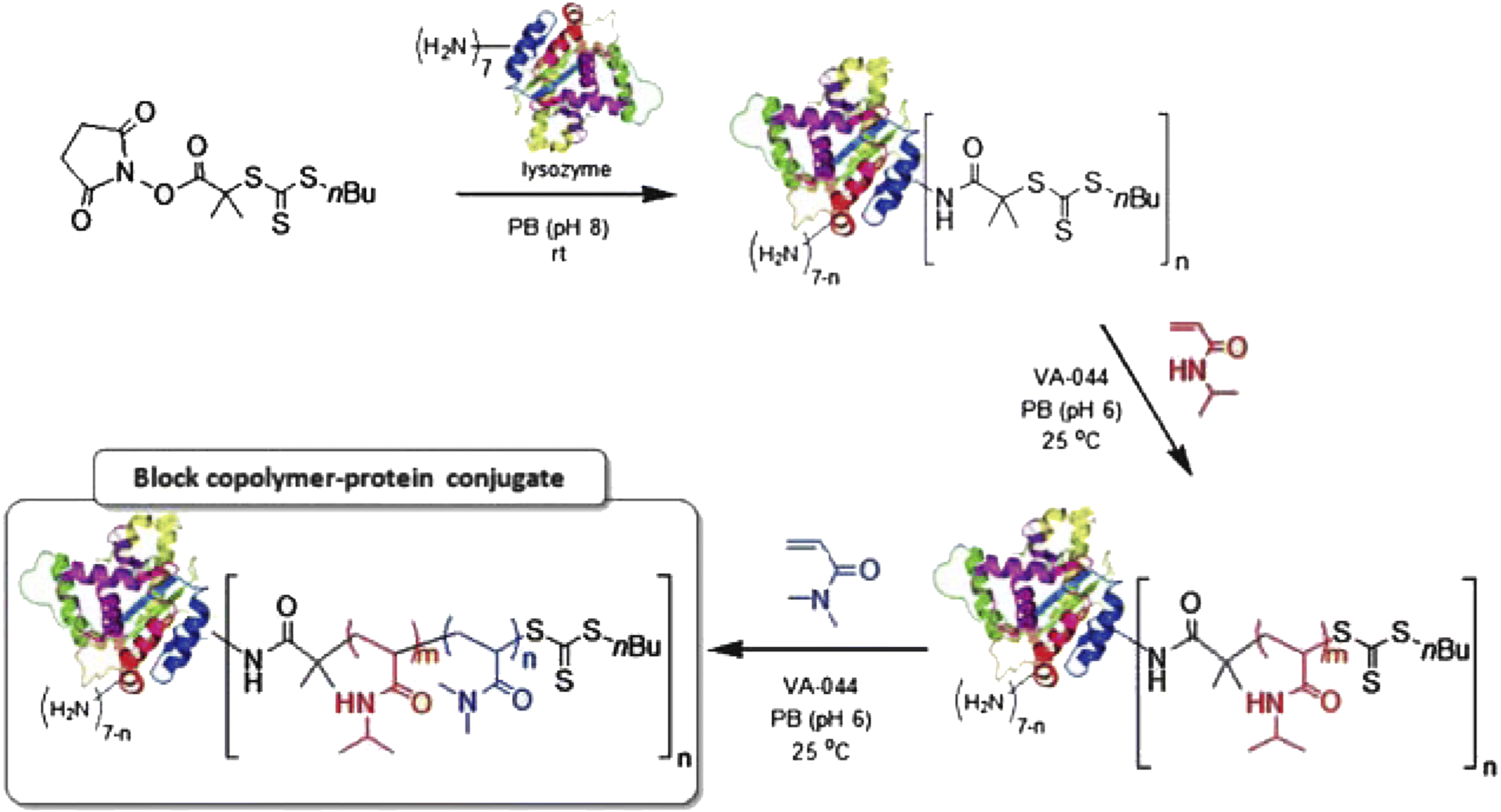




Polym. Chem., **2011**, *2*, 1567

Protein-Polymer Conjugates

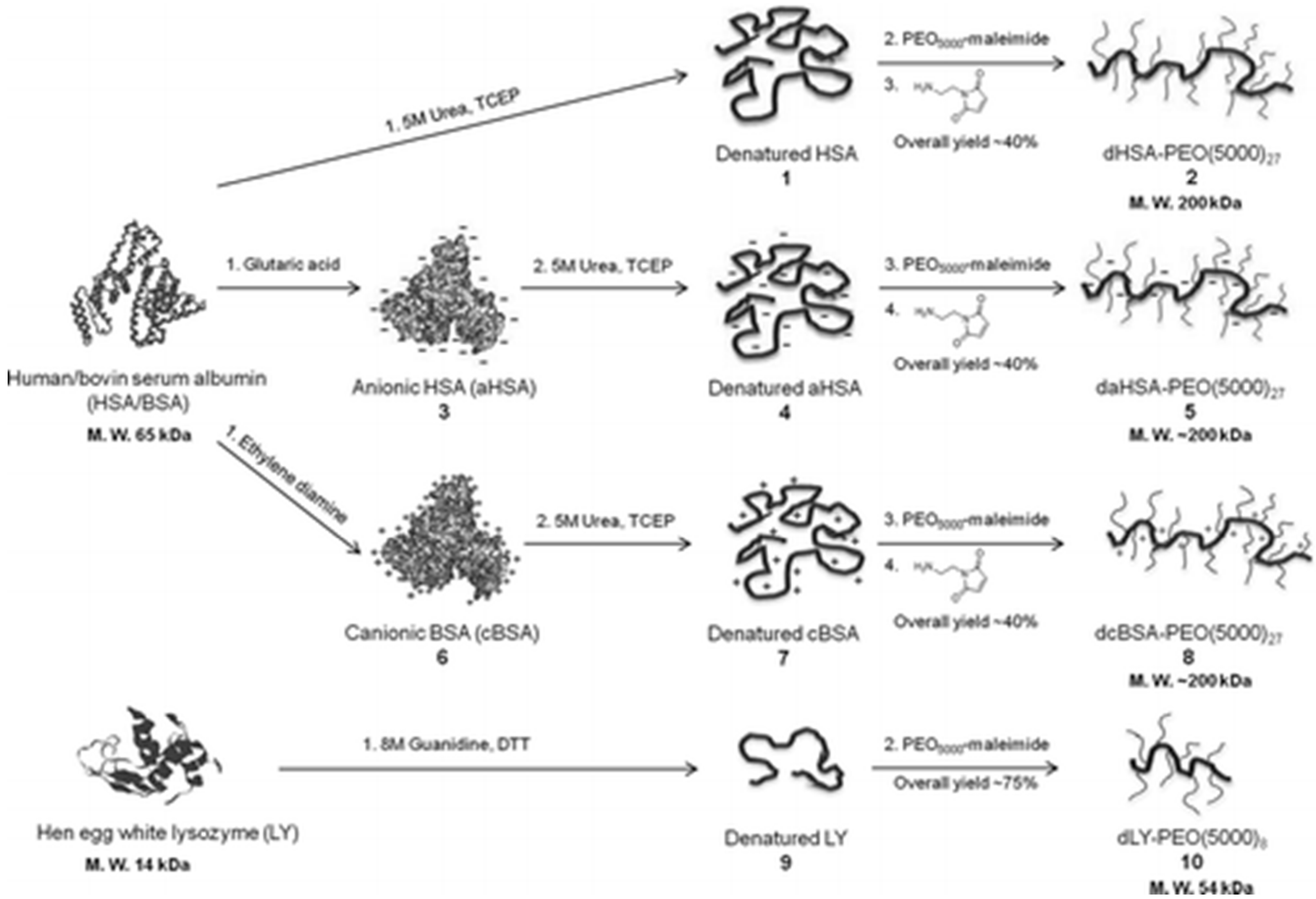
Grafting from



ACS Macro Lett. **2015**, *4*, 101

Protein-Polymer Conjugates

Special Case: Denatured Proteins as Scaffolds



Biomacromolecules **2012**, *13*, 1890

Polymer-Biomolecule Conjugates

- Polymer-Protein Conjugates -

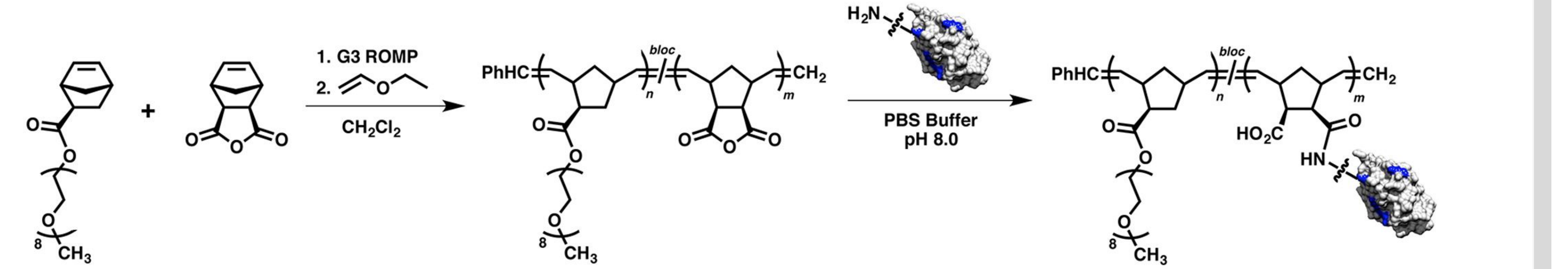
Characterization

Protein-Polymer Conjugates

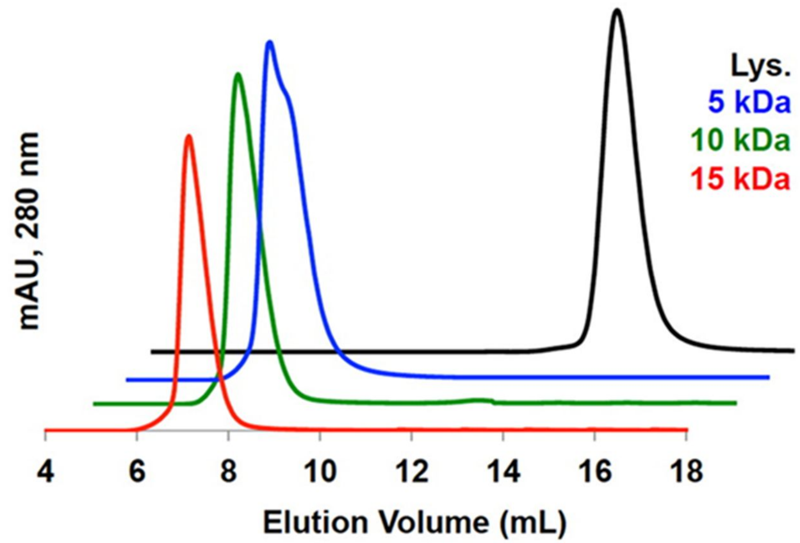
Characterization

Liquid chromatography

SEC: size



PEG-norbornene monomer (1) *PNB block copolymer (2)* *lysozyme:PNB conjugate (3)*



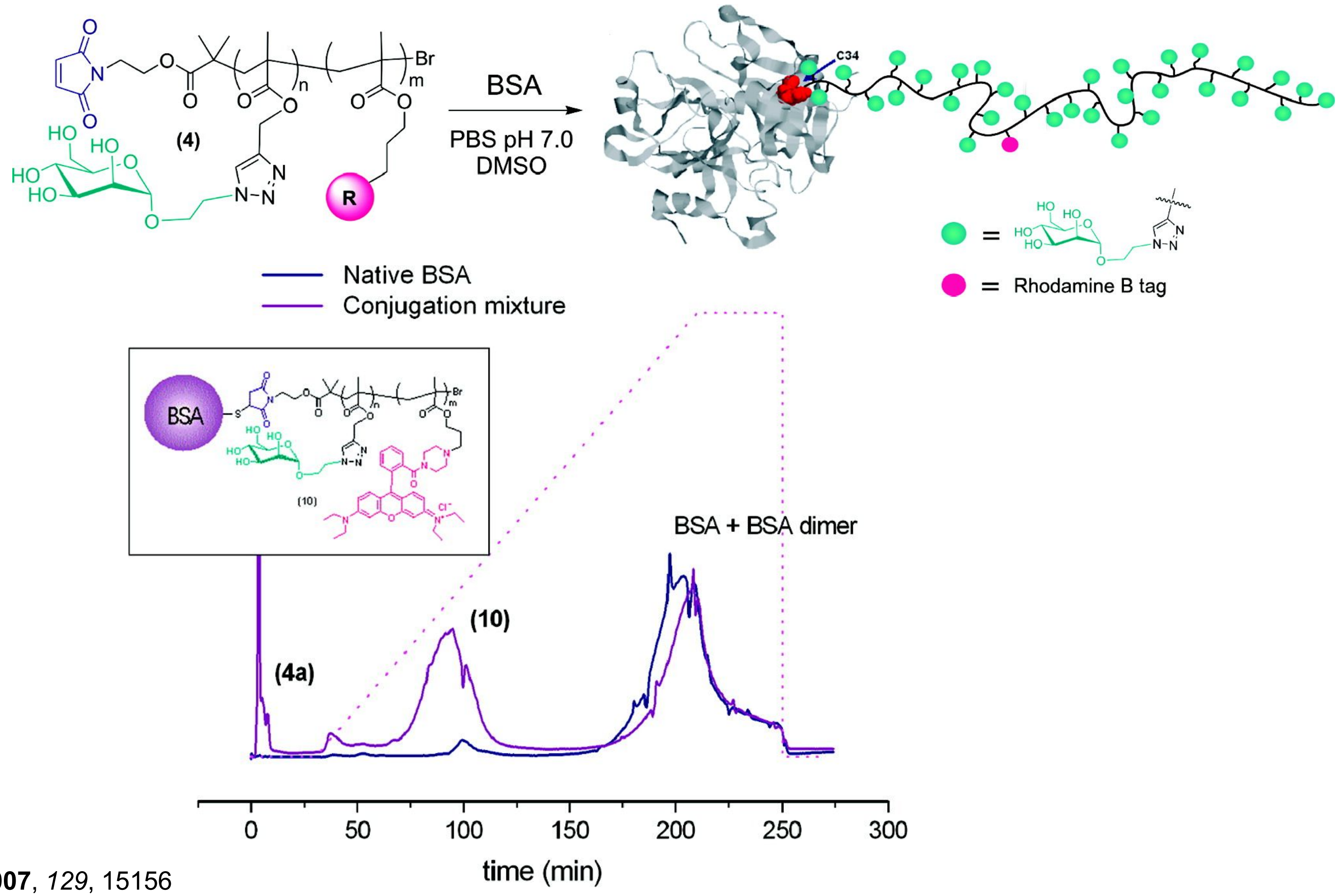
Biomacromolecules **2016**, *17*, 641

Protein-Polymer Conjugates

Characterization

Liquid chromatography

FPLC (fast protein LC): size, ion exchange, hydrophobicity, reverse-phase, or biorecognition

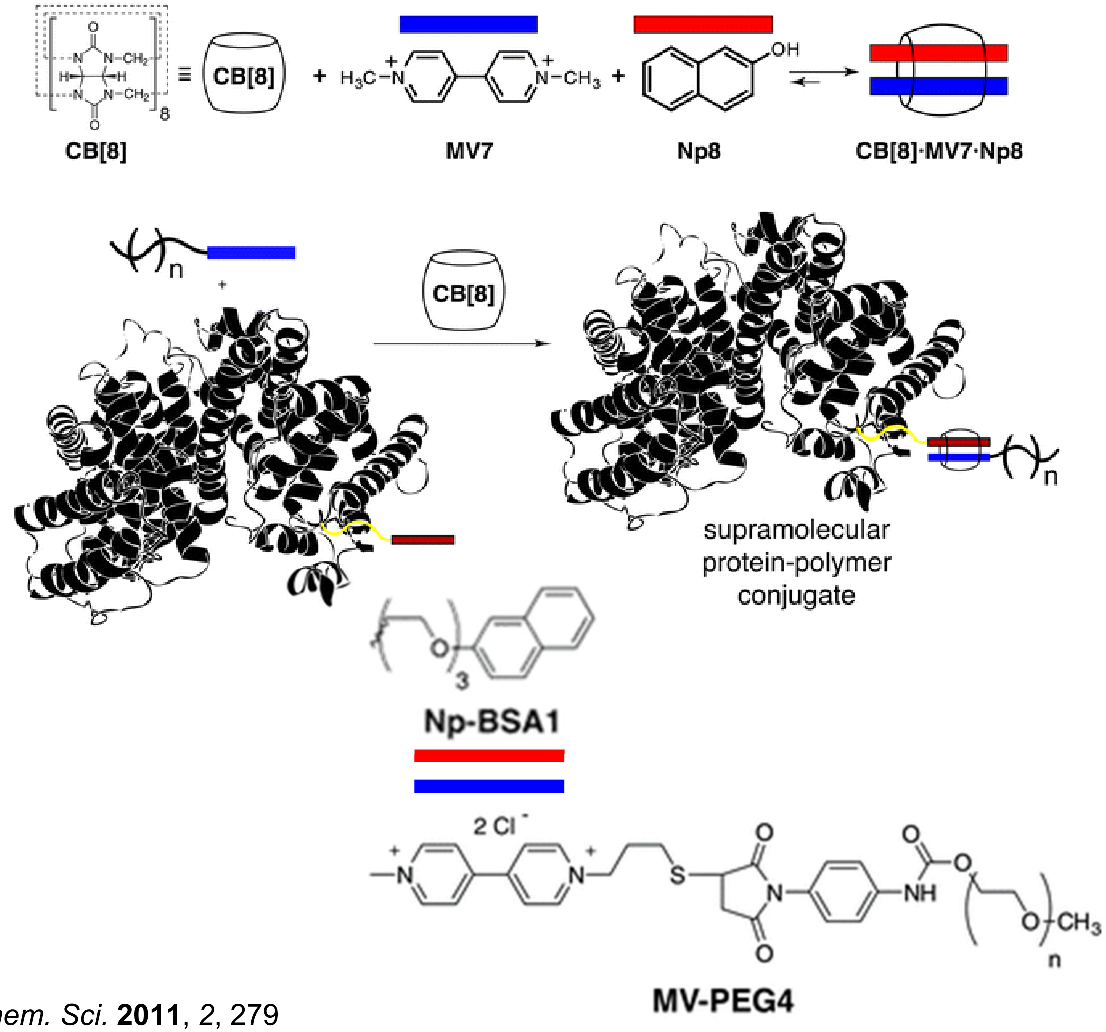


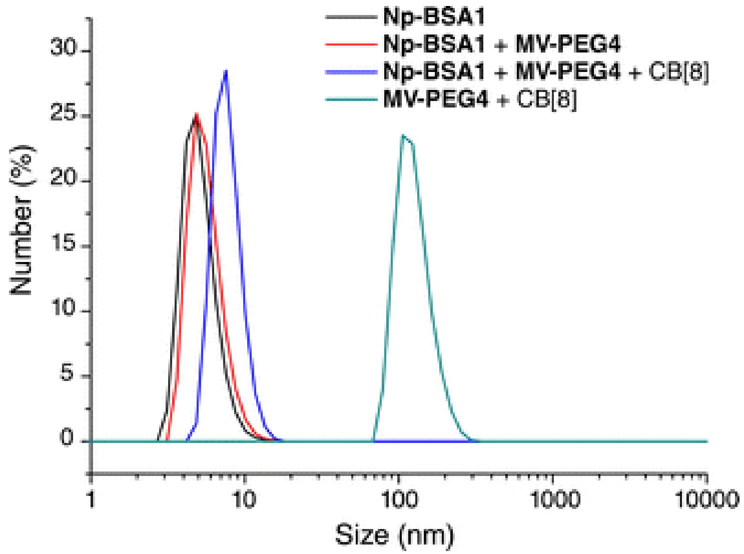
J. Am. Chem. Soc. **2007**, *129*, 15156

Protein-Polymer Conjugates

Characterization

Dynamic light scattering





Chem. Sci. **2011**, 2, 279

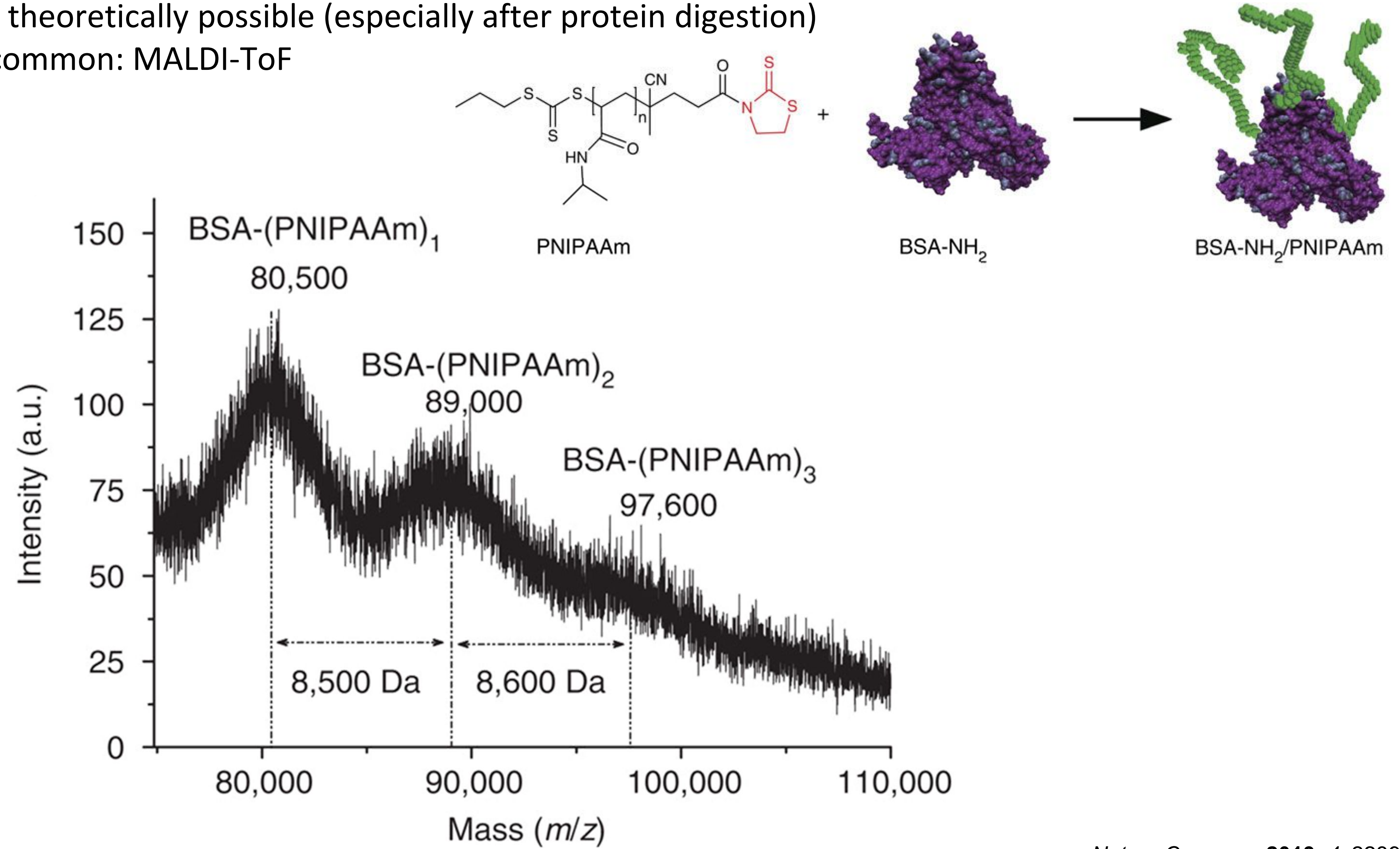
Protein-Polymer Conjugates

Characterization

Mass spectrometry

ESI-MS theoretically possible (especially after protein digestion)

More common: MALDI-ToF



Nature Commun. **2013**, *4*, 2239

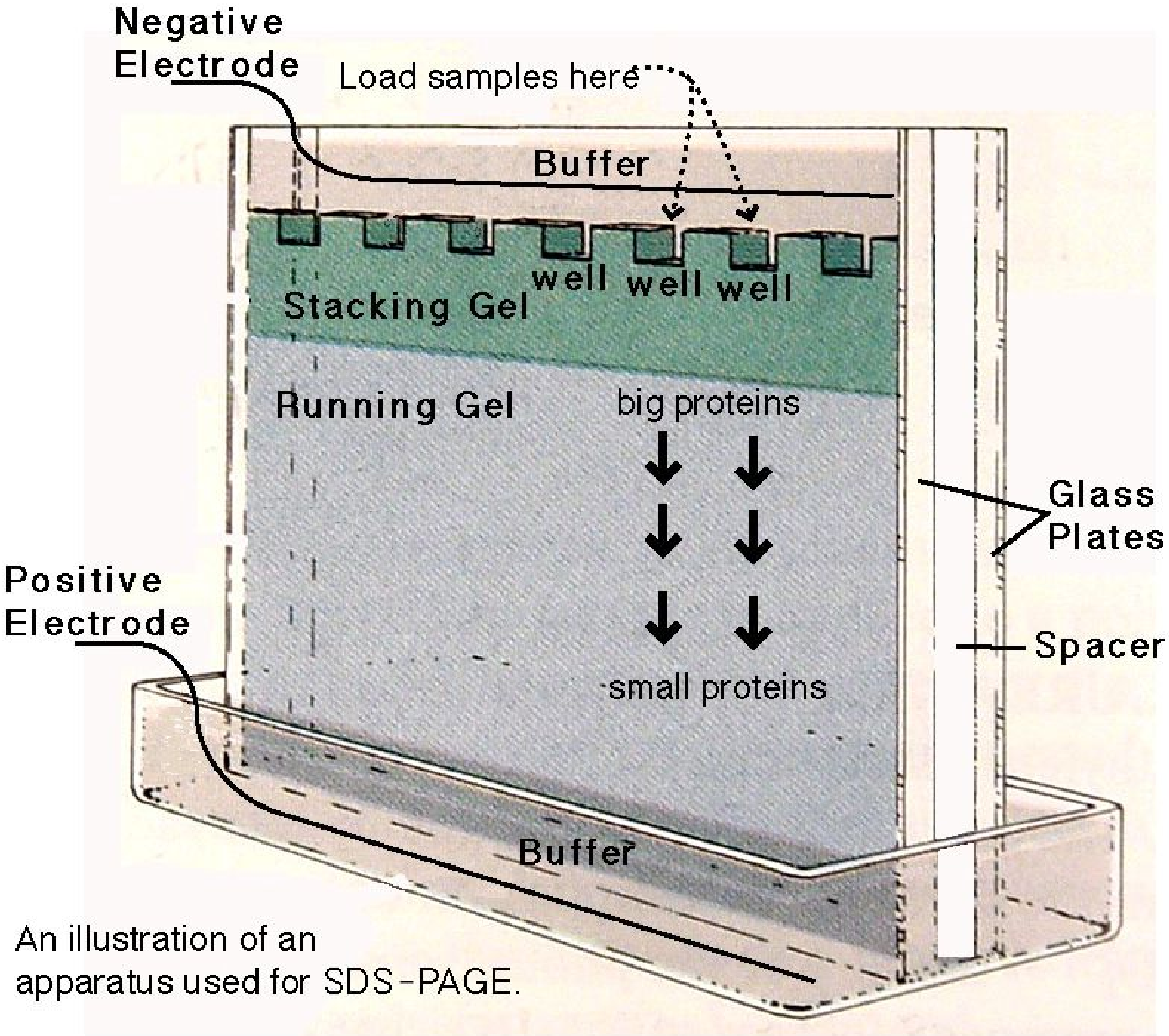
Protein-Polymer Conjugates

Heinrich Heine
University Düsseldorf

Characterization

Gel electrophoresis

Typically SDS-PAGE



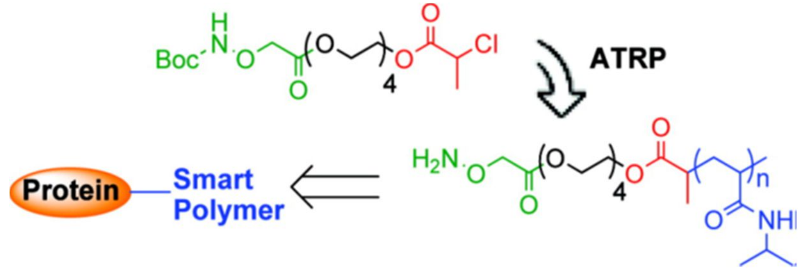
http://www.creative-proteomics.com

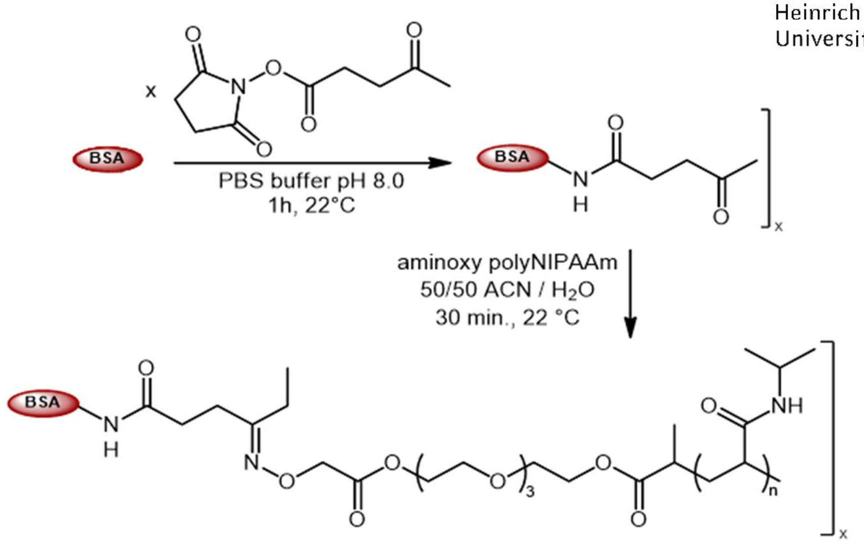
Protein-Polymer Conjugates

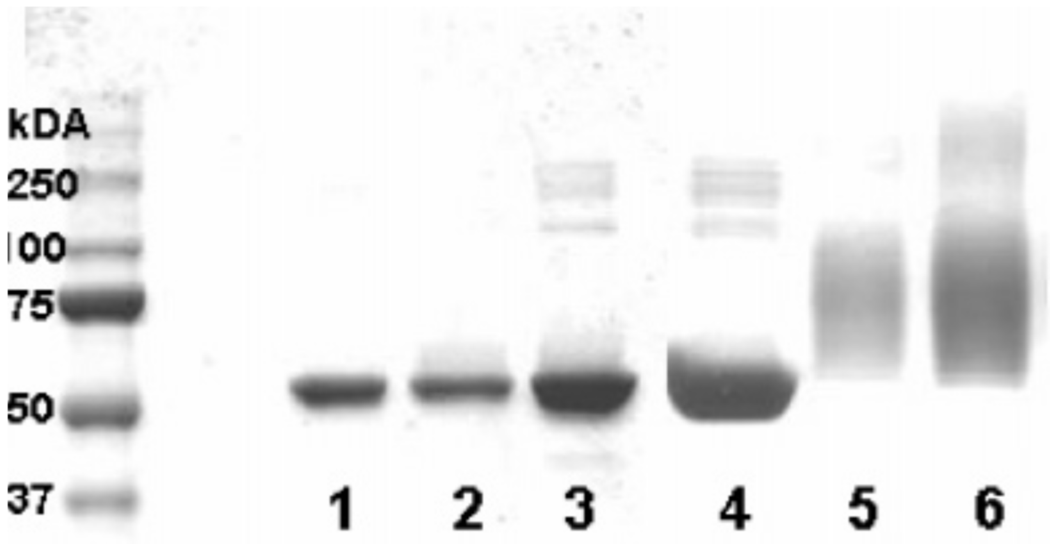
Characterization

Gel electrophoresis

Typically SDS-PAGE







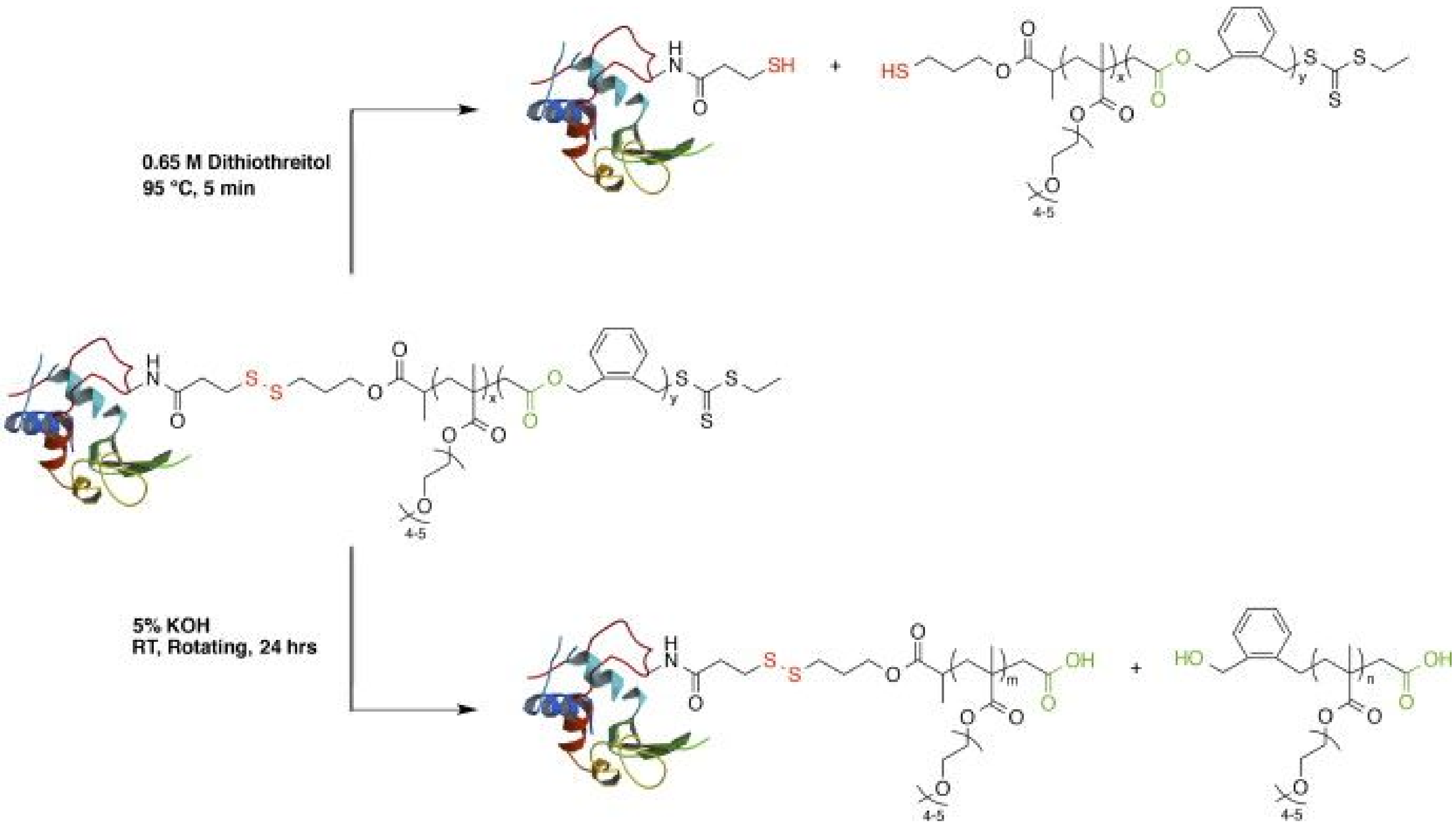
lane 1 aldehyde-BSA;
lane 2 aldehyde-BSA mixed + Boc-NH_2O-NIPAAm
lane 3 unmodified BSA + NH_2O-NIPAAm
lane 4 unmodified BSA
lane 5 polyNIPAAm-BSA crude
lane 6 polyNIPAAm-BSA isolated by thermal precipitation

Protein-Polymer Conjugates

Characterization

Particular cases

Cleavable polymers → SEC!

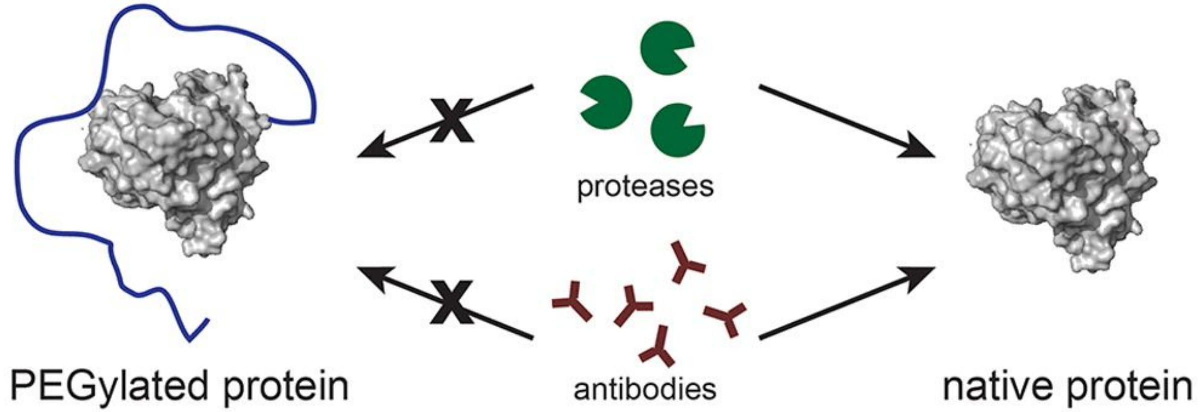


Eur. Polym. J. **2015**, *65*, 305

Polymer-Biomolecule Conjugates
- Polymer-Protein Conjugates -
Applications

Protein-Polymer Conjugates

Protein stabilization/masking



trade name	protein	PEG size	indication	company
adagen	adenosine deaminase	5 kDa	SCID	Enzon
oncaspar	asparaginase	5 kDa	ALL	Enzon
somavert	GH antagonist	5 kDa	excessive growth	Pfizer
PEG-intron	interferon α2b	12 kDa	hepatitis C	Schering-Plough
pegasys	interferon α2a	40 kDa	hepatitis C	Roche
neulasta	granulocyte CSF	20 kDa	neutropenia	Amgen
mircera	EPO	30 kDa	anemia	Roche
cimzia	anti-TNFα Fab’	40 kDa	rheumatoid arthritis	Nektar/UCB
krystexxa	uricase	10 kDa	chronic gout	Savient

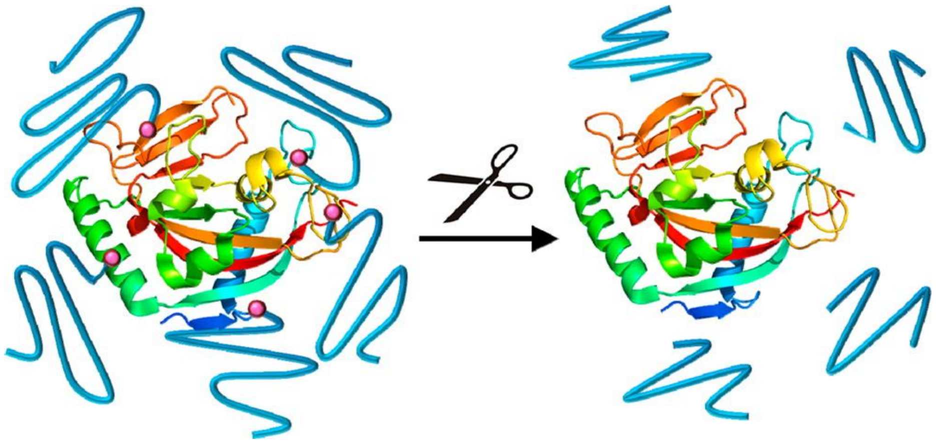
ACS Macro Lett. **2015**, *4*, 101

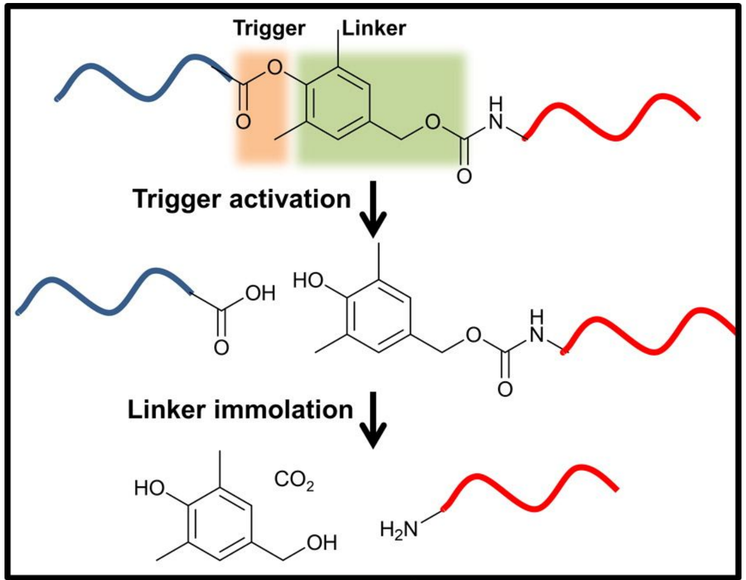
Protein-Polymer Conjugates

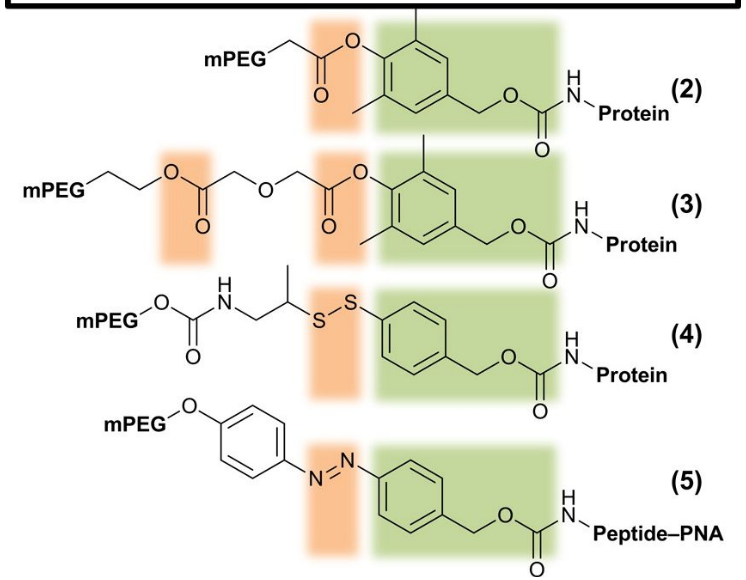
Heinrich Heine University Düsseldorf

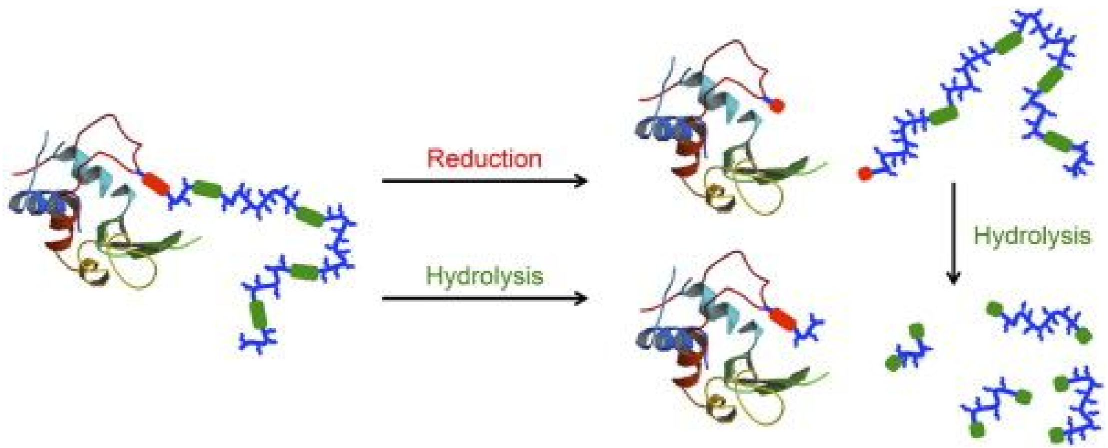
Protein stabilization/masking

Certainly useful for biomedical applications: reversible conjugation





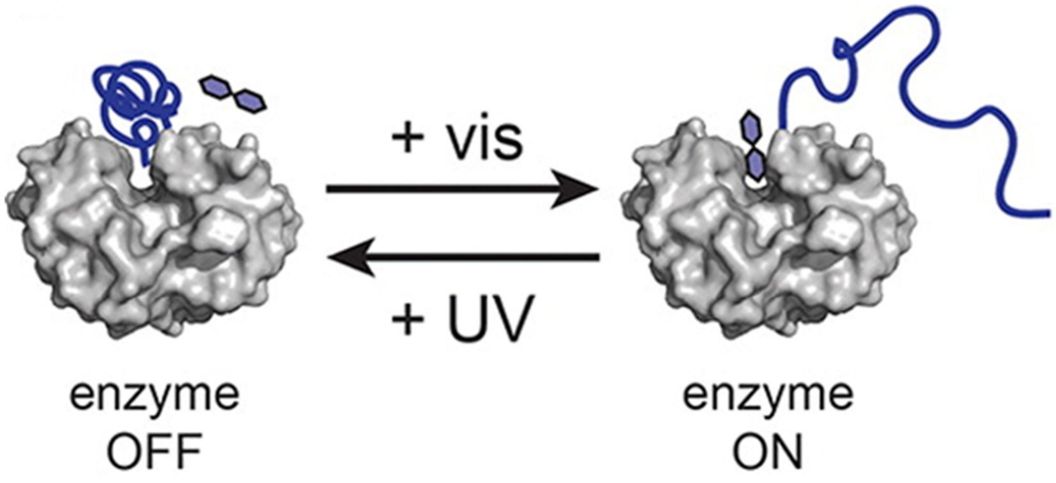


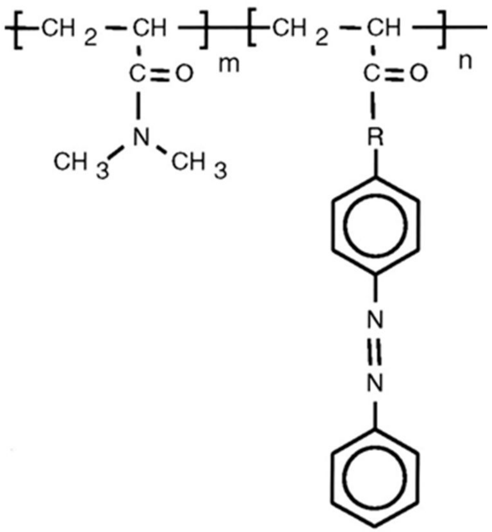


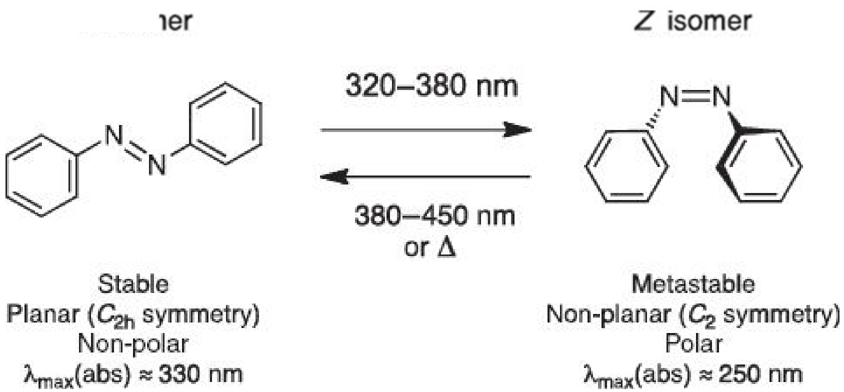
Bioconjug. Chem. **2015**, *26*, 1172; *Eur. Polym. J.* **2015**, *65*, 305;

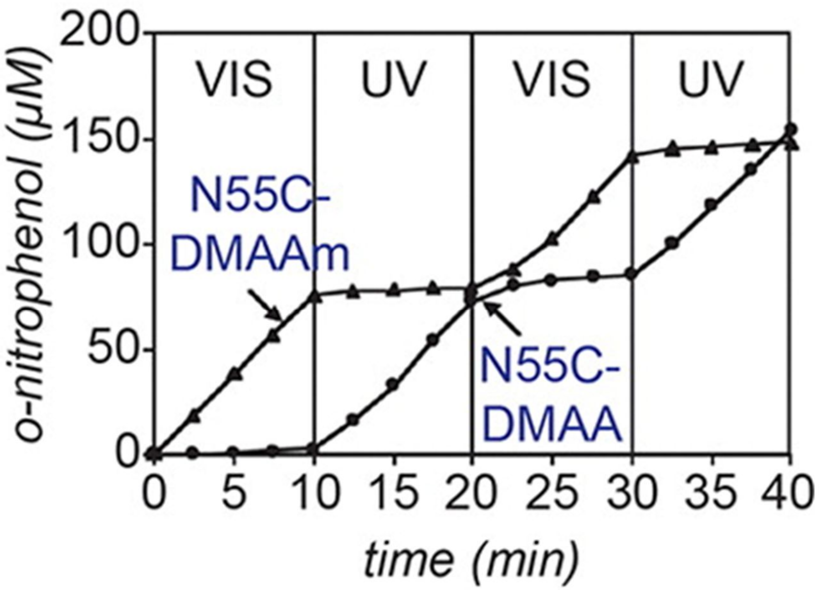
Protein-Polymer Conjugates

Protein activity modulation





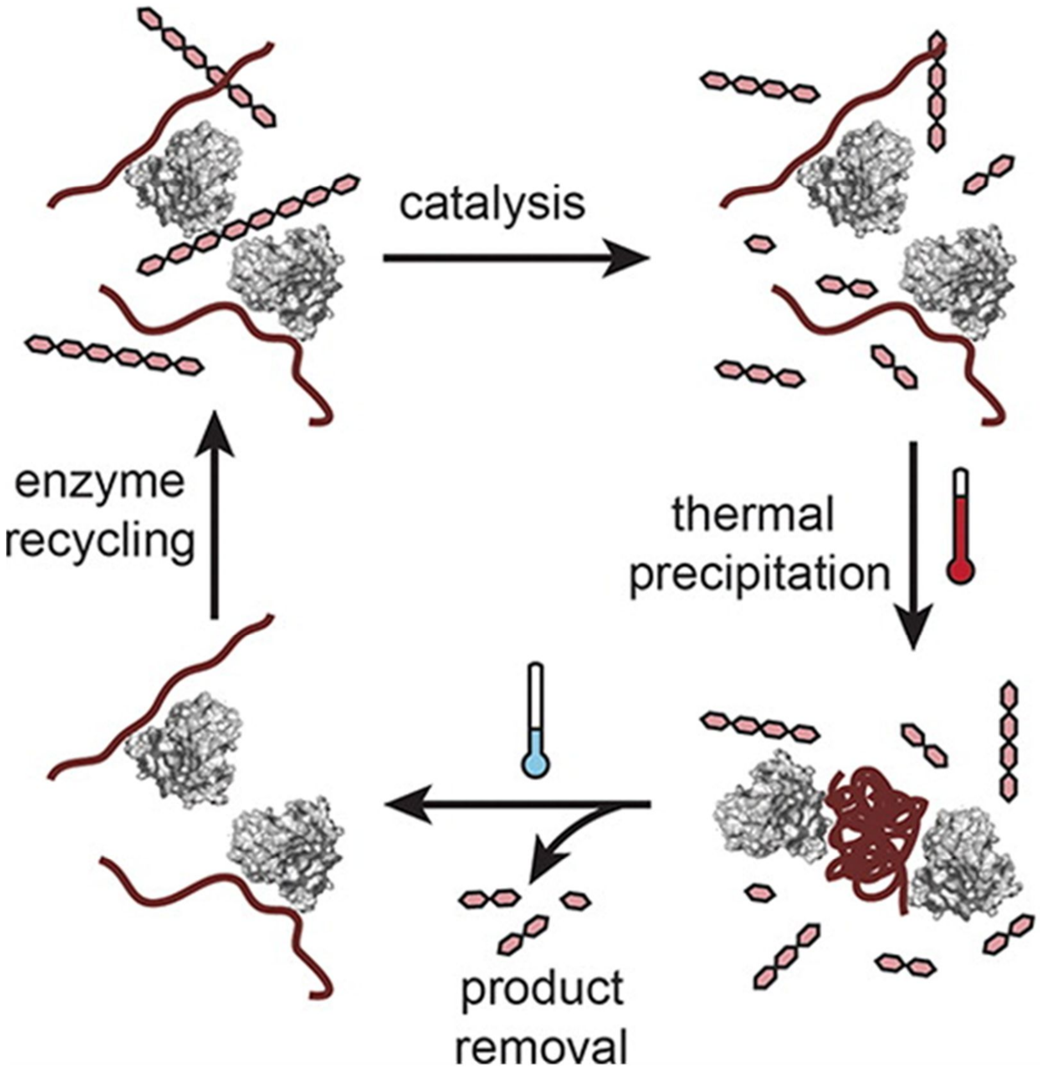


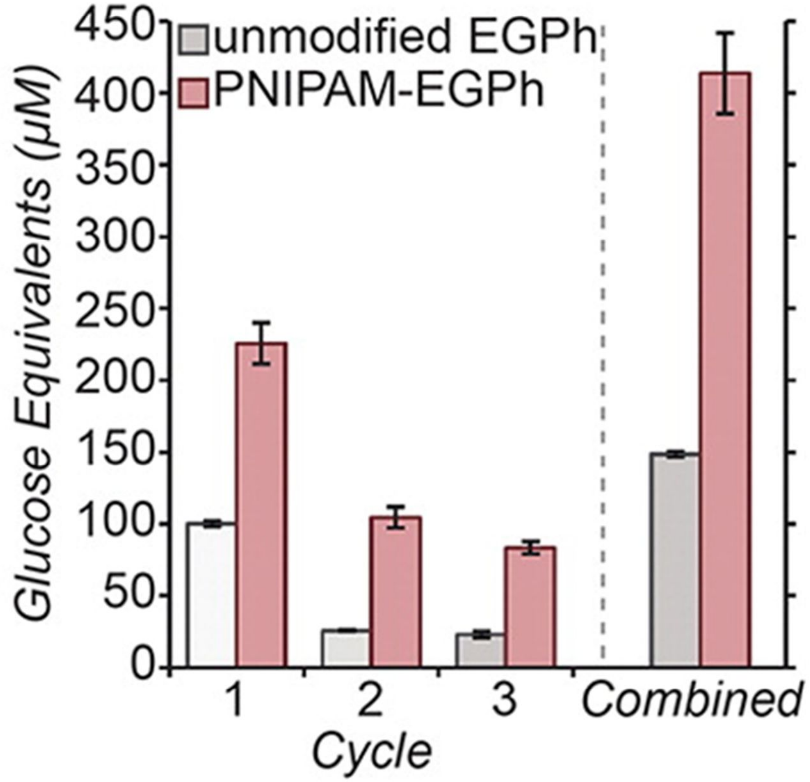


ACS Macro Lett. **2015**, *4*, 101; *Proc. Natl. Acad. Sci. U.S.A.* **2002**, *99*, 16592

Protein-Polymer Conjugates

Protein recovery

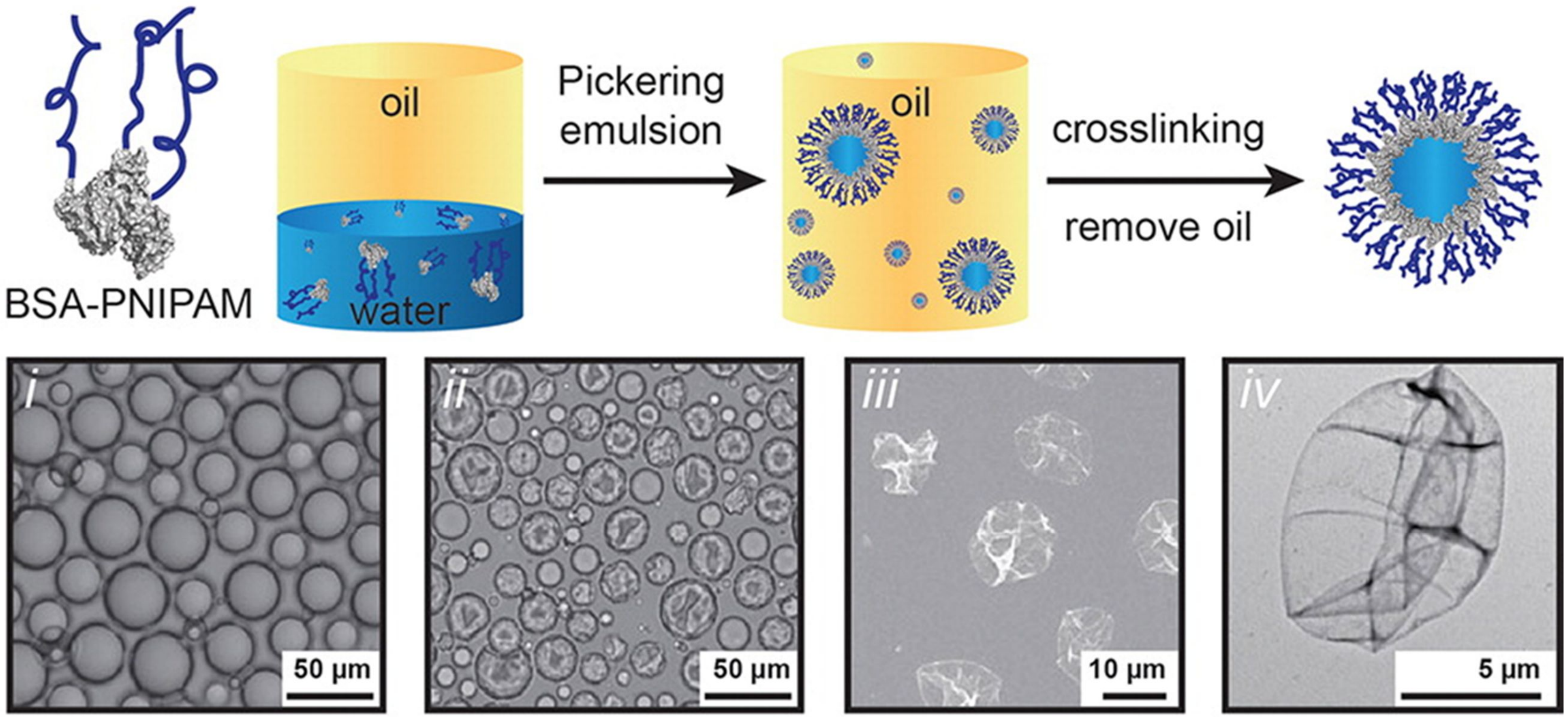




ACS Macro Lett. **2015**, *4*, 101

Protein-Polymer Conjugates

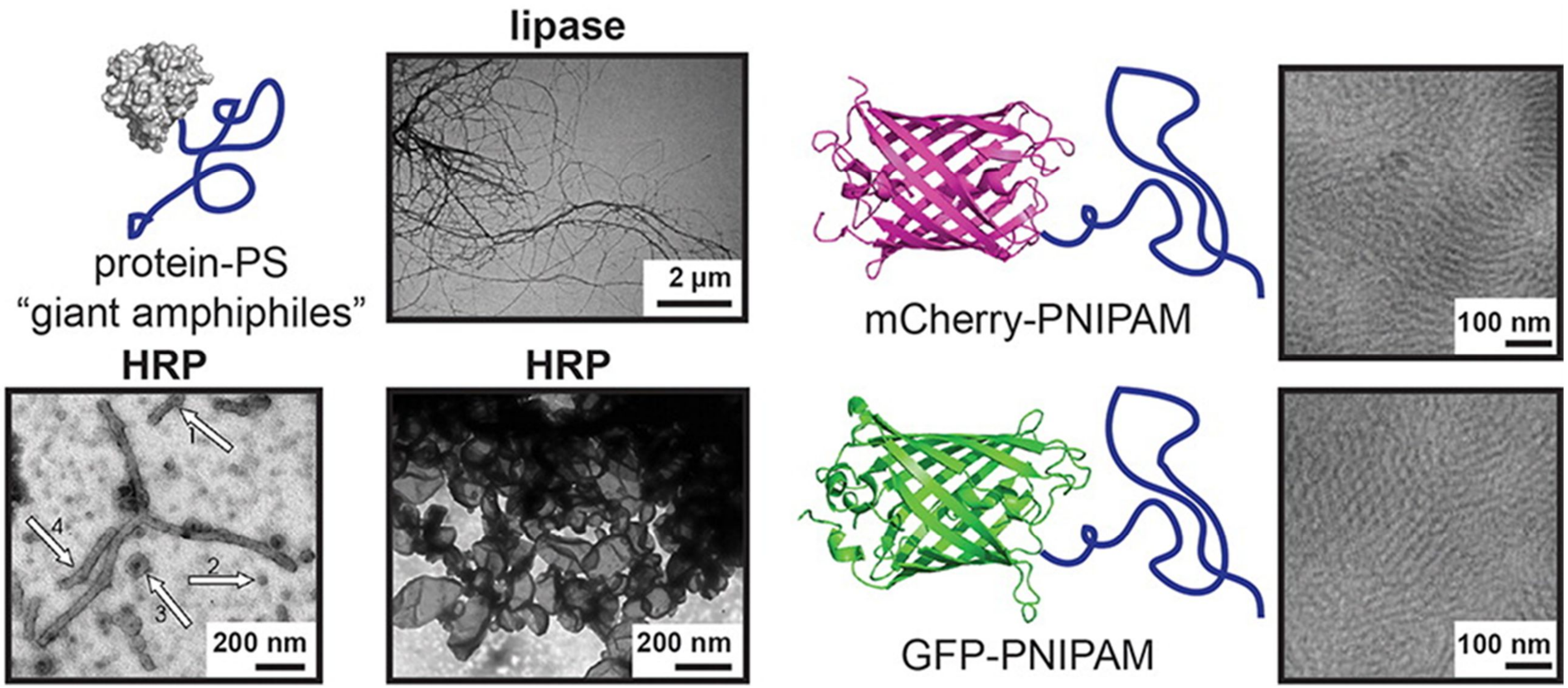
Nanocontainers/nanoreactors



ACS Macro Lett. **2015**, *4*, 101

Protein-Polymer Conjugates

Structural materials (expensive!)



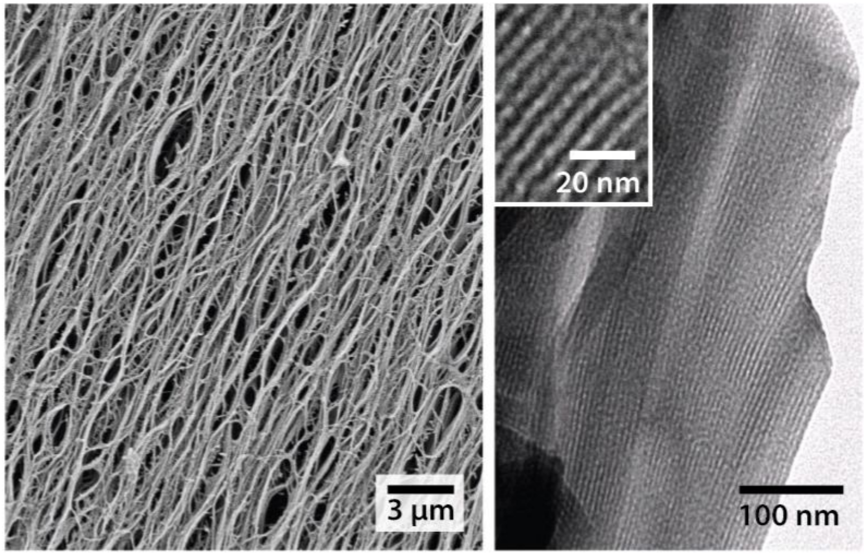
ACS Macro Lett. **2015**, *4*, 101

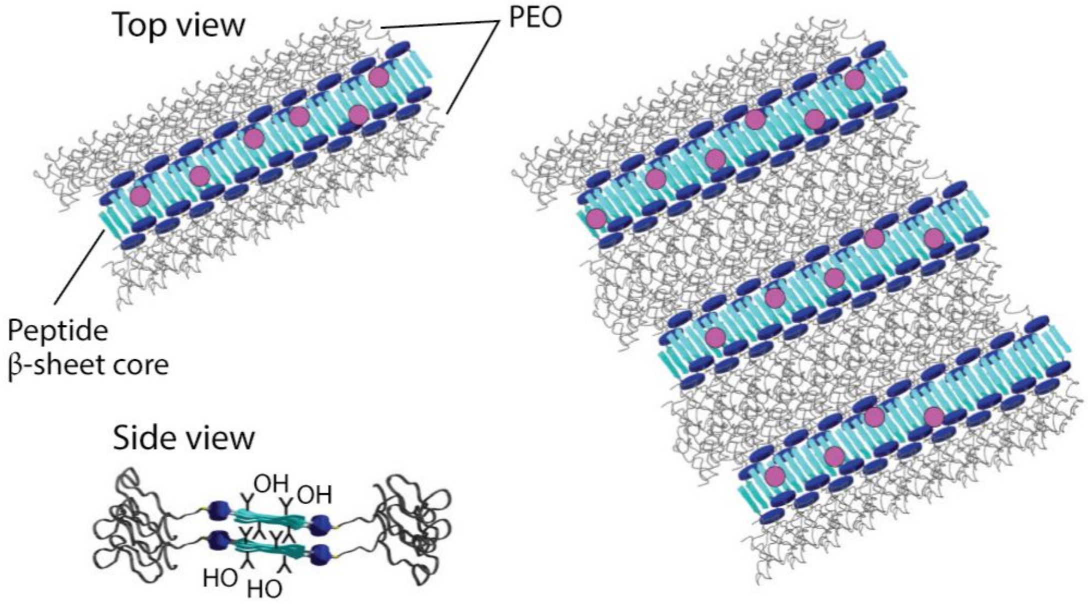
Polymer-Biomolecule Conjugates
- Other examples -

Peptides

Similar to proteins
basically a simpler version

Polymer conjugation can also be achieved on solid phase



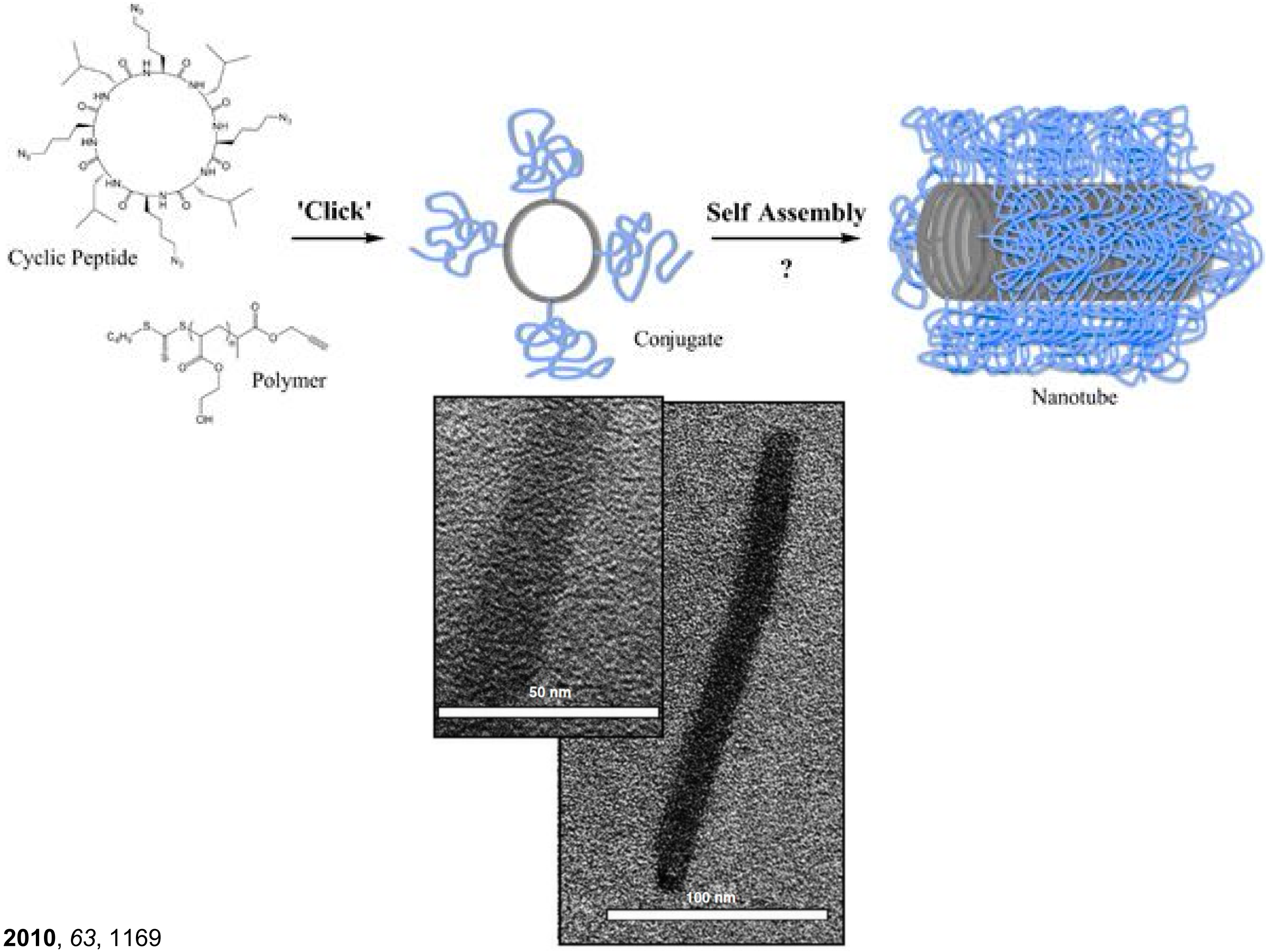


Annu. Rev. Phys. Chem. **2013**, *64*, 631

Peptides

Similar to proteins

basically a simpler version

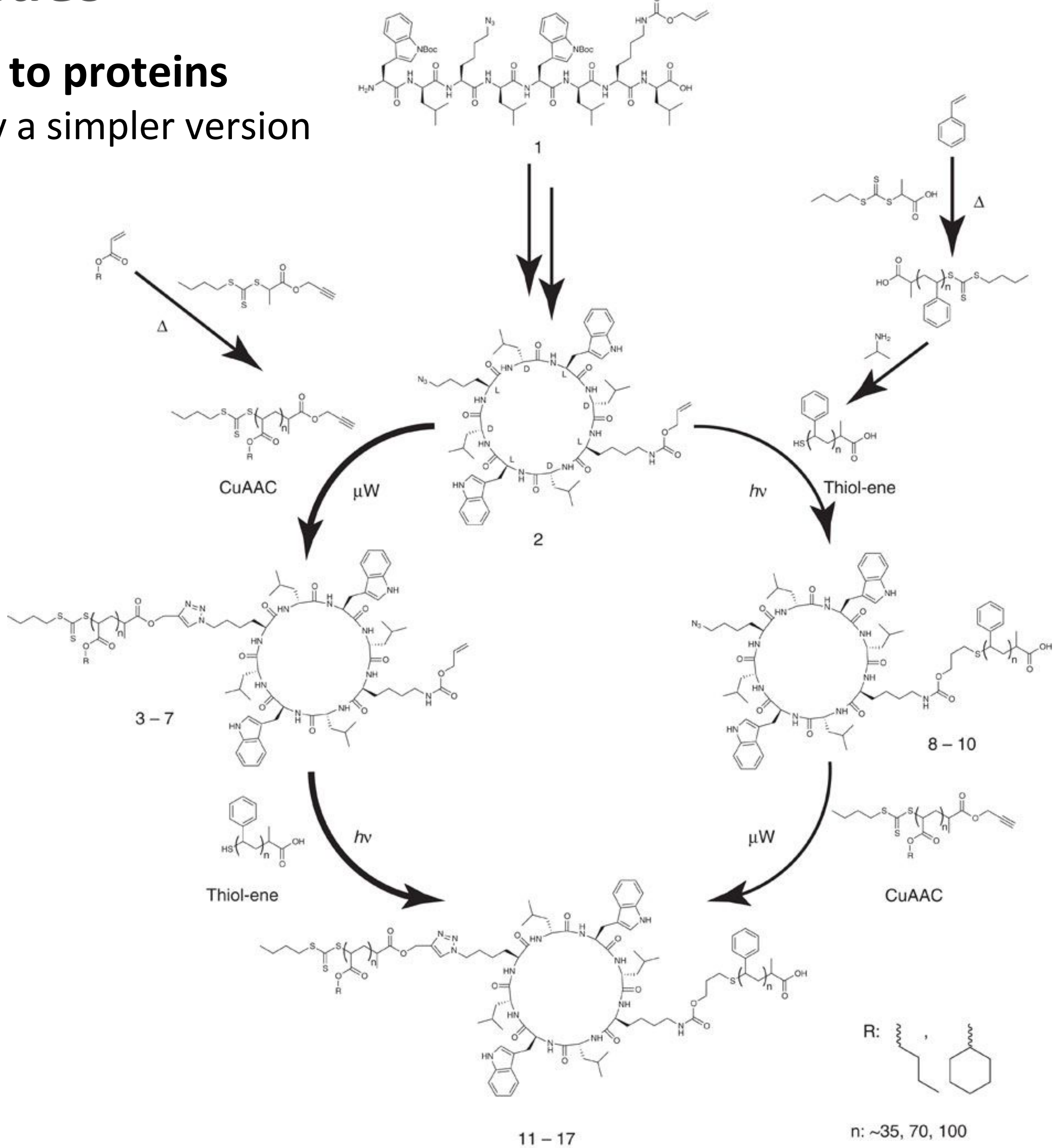


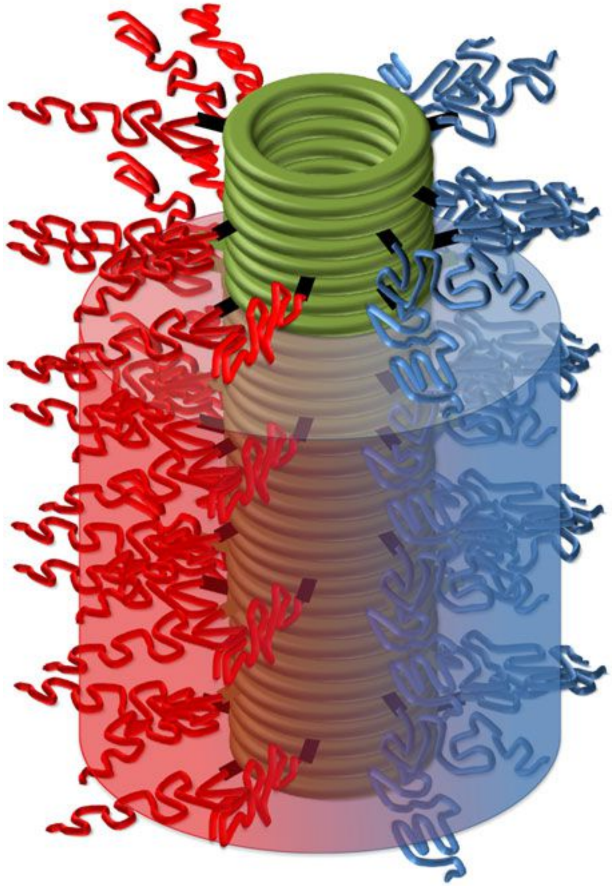
Aus. J. Chem. **2010**, *63*, 1169

Peptides

Similar to proteins

basically a simpler version

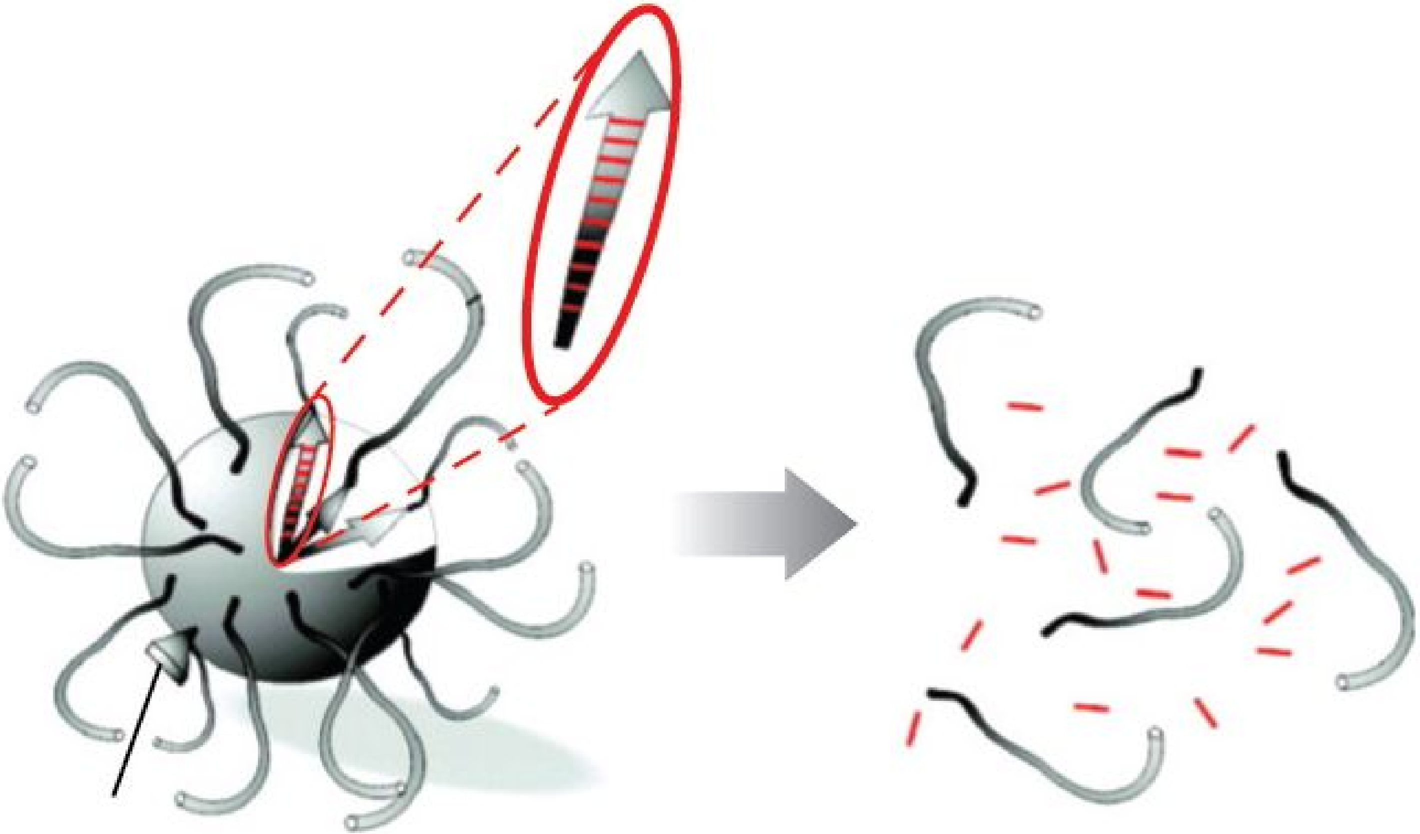




Nature Commun. **2013**, *4*, 2780

Peptides

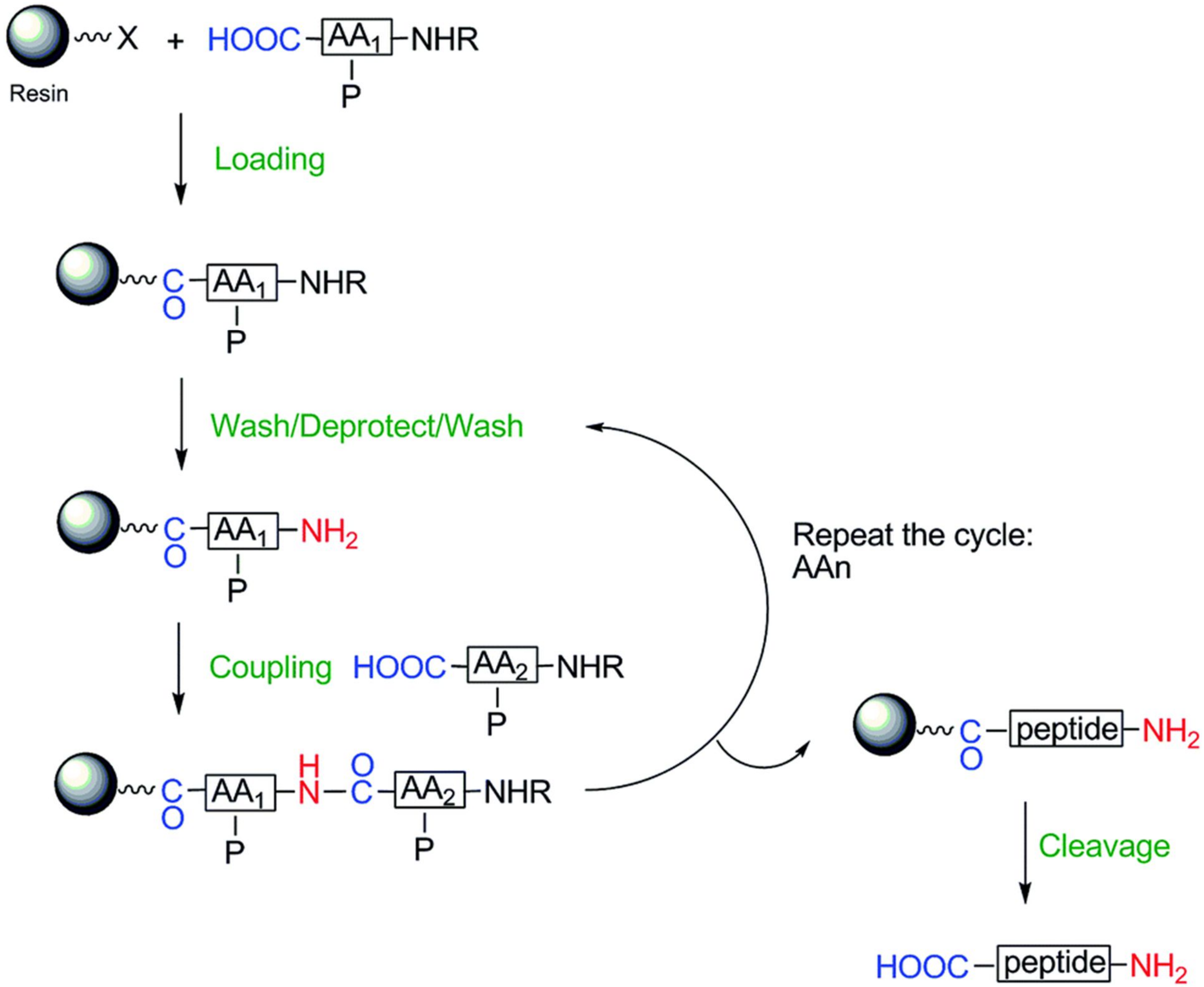
Similar to proteins
basically a simpler version



Annu. Rev. Phys. Chem. **2013**, *64*, 631

Peptides

Peptide Synthesis – *Solid-phase synthesis*

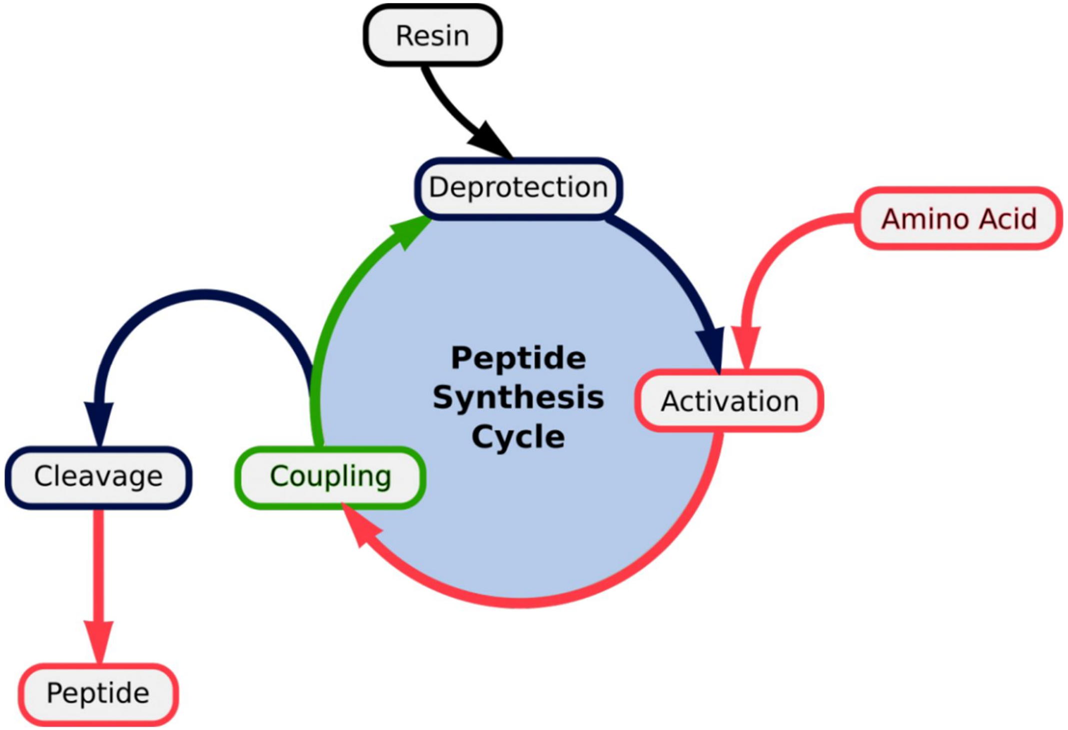


Just one example!

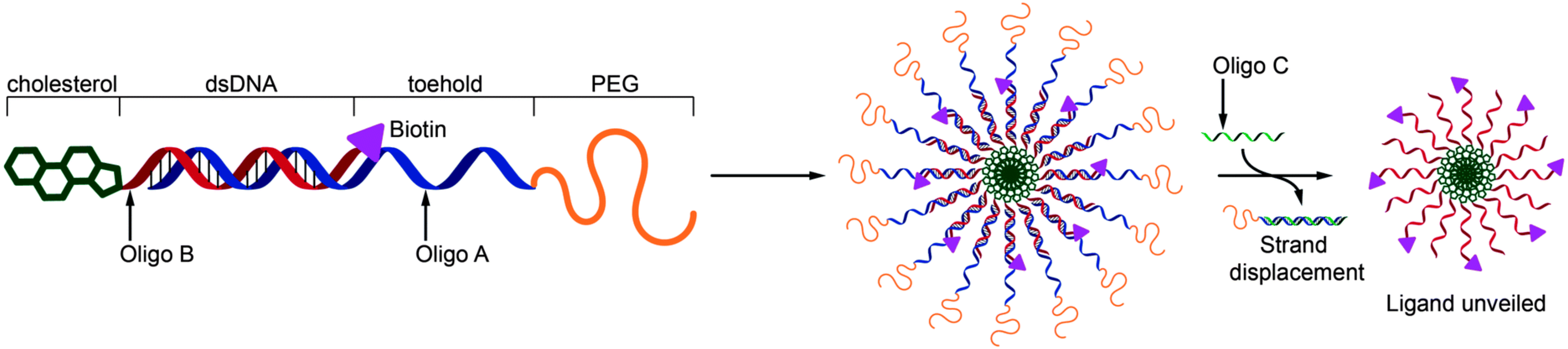
RSC Adv. **2014**, *4*, 32658

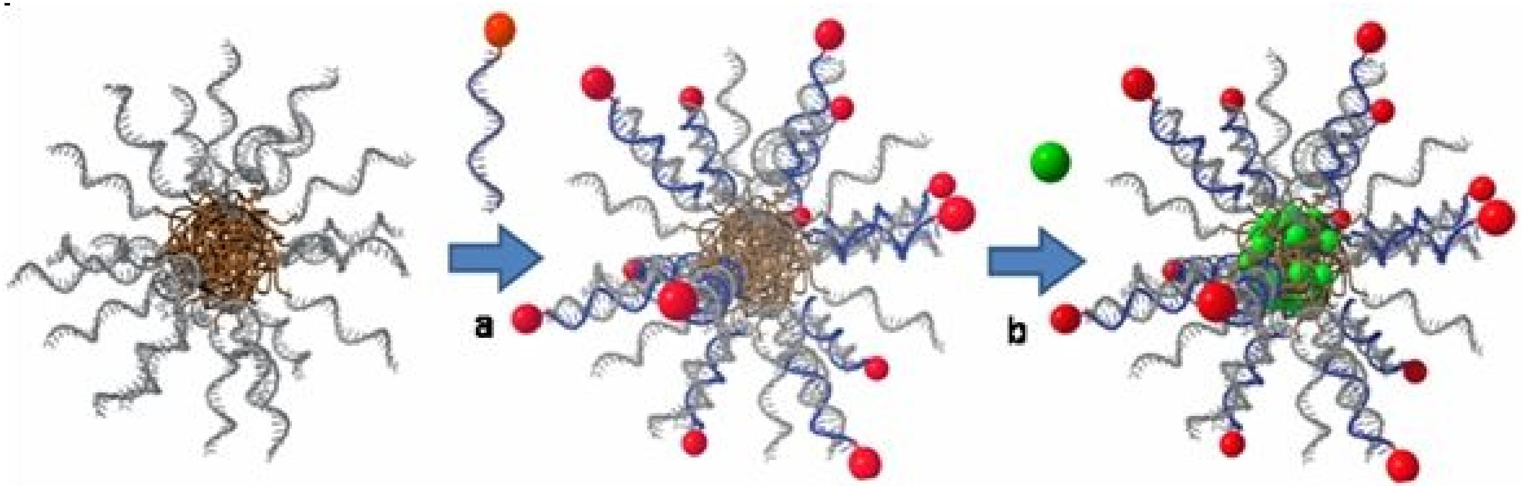
Peptides

Peptide Synthesis – *Solid-phase synthesis*



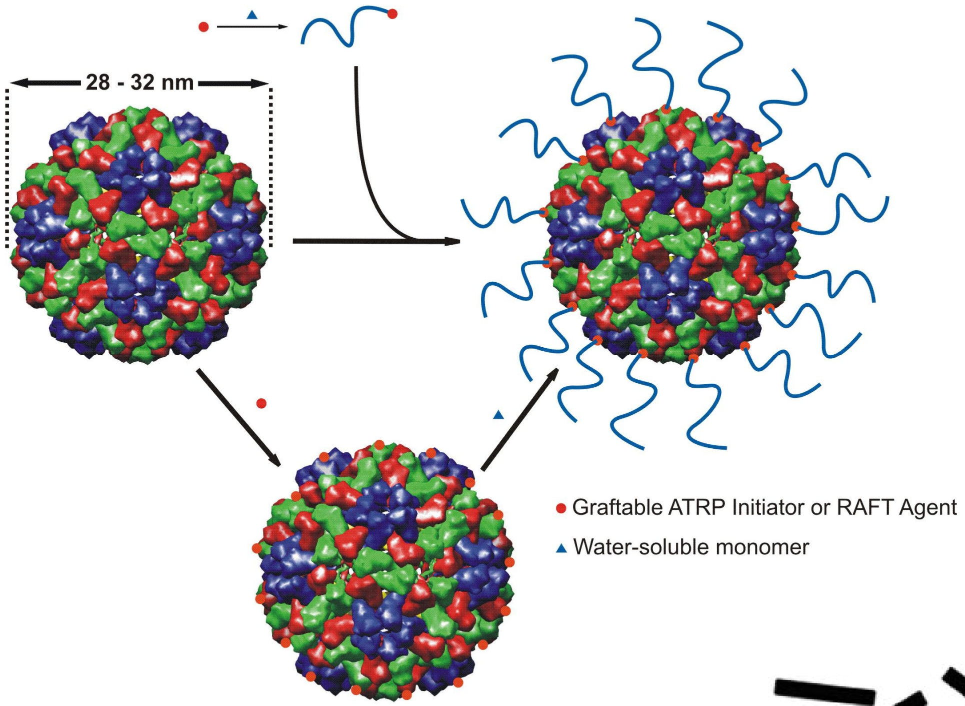
DNA/RNA

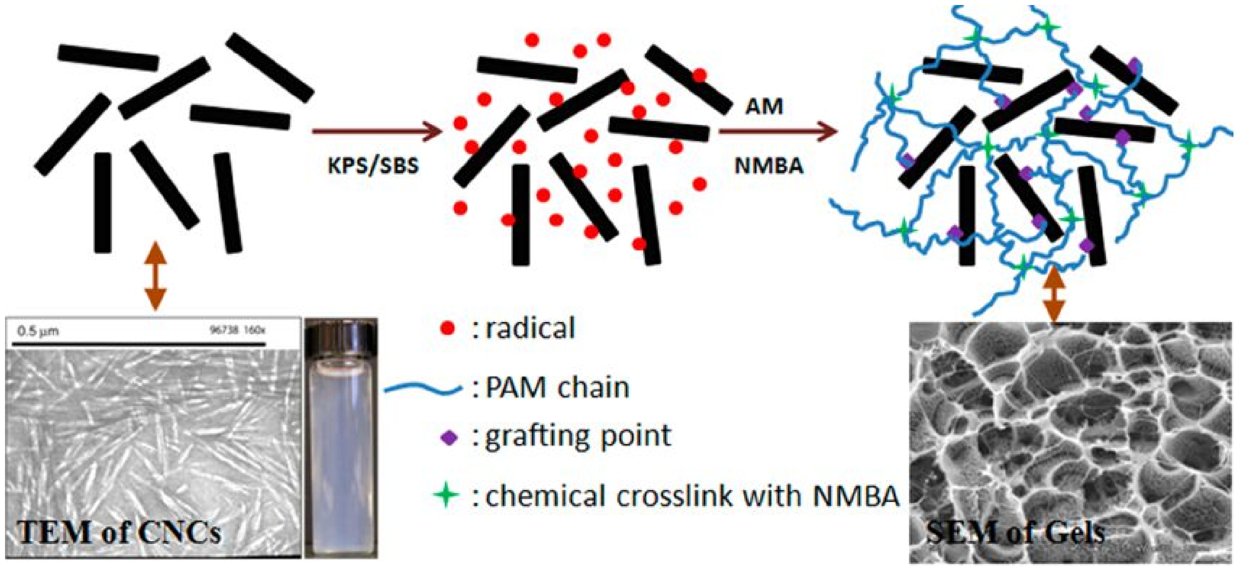




Nanoscale **2014**, *6*, 2368; *Angew. Chem. Int. Ed.* **2010**, *46*, 8574

Viruses, Polysaccharides…





Vielen Dank!